

Permansson Regimes: A General Framework for Strategic Dynamics Beyond Equilibrium

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Abstract

Game-theoretic and learning models often produce a behavioral process whose long-run consequences are then analyzed with tools from dynamical systems, stochastic approximation, occupation measures, or viability theory. The earlier Equilibrium-Generated Regime (EGR) framework formalized a narrower post-equilibrium object: a certified equilibrium policy together with an ex ante regime region, basin, persistence requirement, descriptor, limiting occupation law, convergence mode, and intervention-relative constitutive test. This paper removes equilibrium as a primitive of the regime layer while preserving the strategic semantics that motivated EGR. A canonical strategic-world process separates an action-selection kernel, a world-transition kernel, and a strategic-update kernel; their composition induces an ordinary Markov process on the enlarged state space. Generated Regimes (GRs) are ex ante classifications of that induced process, while generalized Permansson Regimes (PRs) are GRs with at least one strategically constitutive generator component under a fixed admissible intervention. The paper proves well-posedness, exact EGR embedding, conservative recovery of the original Permansson definition, baseline representation invariance, factorization non-identification, and constitutive non-invariance under baseline-equivalent factorizations.

The framework further develops a quantitative and representation-theoretic layer. Descriptor spaces may be Polish; finite persistence and expected exit time admit an exact killed-kernel calculus; finite-horizon quasi-regimes may carry an additional quasi-stationary-distribution (QSD) certificate subject to separate existence and uniqueness conditions; constitutive effects admit a uniform margin and a perturbation-robustness criterion; intervention-compatible invariance extends to frozen intervention families; and a sufficient quotient theorem gives conditions under which GR/PR classification survives type-respecting state compression. These results are modular and apply only under their stated hypotheses. Formal verification and reproducibility materials are documented in Appendix D.

Keywords: generated regimes; Permansson regimes; dynamic games; learning in games; constitutive intervention; killed Markov kernel; quasi-stationary distribution; perturbation robustness; regime equivalence; probabilistic bisimulation.

JEL codes: C72, C73, D02, D74.

1. Introduction

1.1 The analytical question

A strategic model may identify an equilibrium, a learning rule, an evolutionary revision protocol, a bounded-rational adjustment process, or a fixed policy. Once such a process is specified, a second question remains: what persistent or structurally meaningful world does repeated strategic behavior generate? In static game theory the natural endpoint is a solution concept. In dynamic game theory the strategic solution is coupled to a state process. In evolutionary and learning models the behavior itself may remain in motion. Across these settings, the world altered by behavior can in turn reshape future incentives, information, feasible actions, or strategic updates.

The earlier Equilibrium-Generated Regime framework was built for the special case in which the strategic behavior was already certified by an independently established dynamic equilibrium. It deliberately did not introduce a new equilibrium concept. Instead, it attached a post-equilibrium classification object to the equilibrium-induced process: a selected equilibrium policy, an ex ante region and initial basin, exact persistence, a descriptor, a limiting occupation law, a convergence mode, and an intervention-relative test for whether a component of equilibrium behavior was constitutive of a defining regime property (Hermansson, 2026a).

The present paper asks whether equilibrium was ever logically necessary to that regime-classification layer. The answer developed here is no, but only in a specific and conservative sense. The mathematics of non-equilibrium dynamics is not new: dynamical systems theory classifies invariant sets, attractors, recurrence, chain recurrence, and related asymptotic objects; stochastic approximation connects learning algorithms to differential equations or inclusions; controlled Markov process theory uses occupation measures; viability theory studies controlled persistence; evolutionary game theory separates games from revision protocols; and feedback-evolving games explicitly couple strategic behavior to environmental states. The proposed contribution is therefore not a new theory of dynamics. It is a typed, game-facing regime semantics that preserves the distinction between how strategic behavior is generated, how that behavior changes the world, and which typed components are constitutive of a declared regime property.

1.2 Main contribution

The paper introduces a canonical discrete-time strategic-world process with three kernels. An action-selection kernel α generates behavior from the current strategic and world states. A world-transition kernel P maps current state and behavior into the next world state. A strategic-update kernel U maps the resulting observation back into the next strategic state. The pair $\mathfrak{G} = (\alpha, U)$ is called the strategic generator. Their composition induces an ordinary Markov kernel $K_{\mathfrak{G}, P}$ on the joint state $Y = S \times X$.

$$Y_t = (S_t, X_t) \xrightarrow{\alpha} A_t \xrightarrow{P} X_{t+1} \xrightarrow{U} S_{t+1}. \quad (1)$$

A Generated Regime (GR) is then an ex ante regime specification satisfied by the induced joint process. A generalized Permansson Regime (PR) is a GR for which at least one predeclared component of the strategic generator is constitutive of a defining regime property under a fixed admissible strategic-generator intervention. Structural constitution of the world-transition mechanism is defined separately. For confirmatory use, the framework additionally requires a nondegenerate baseline specification and a grounded constitutive property tied to a frozen relevance map; these guardrails exclude descriptor variation that exists only on unreachable decorative states and PR claims driven only by undeclared auxiliary bookkeeping coordinates. This preserves the logic of the earlier EGR paper, in which a Permansson regime was narrower than an EGR rather than synonymous with it.

The quantitative and representation-theoretic layer sharpens this architecture in five directions. First, it weakens the descriptor-space assumption from compact metrizable to Polish while preserving the weak-convergence formulation. Second, it replaces the one-step finite-survival diagnostic by an exact killed-kernel calculus, so finite persistence and expected exit time become operator identities rather than only lower bounds. Third, it separates constitutive status from a quantitative constitutive margin and proves a metric perturbation theorem showing exactly when that margin survives model or numerical error. Fourth, it generalizes single-intervention representation invariance to a frozen, indexed intervention-family signature. Fifth, it gives a sufficient quotient theorem: type-respecting state compression preserves GR/PR classification only when both models retain the canonical typed structure, the baseline kernel and every frozen typed intervention kernel intertwine, and the regime/constitutive semantics descend together. None of these results makes perturbation robustness, QSD structure, or quotientability automatic; each applies only under explicit hypotheses.

Formal verification scope and reproducibility provenance are documented in Appendix D.

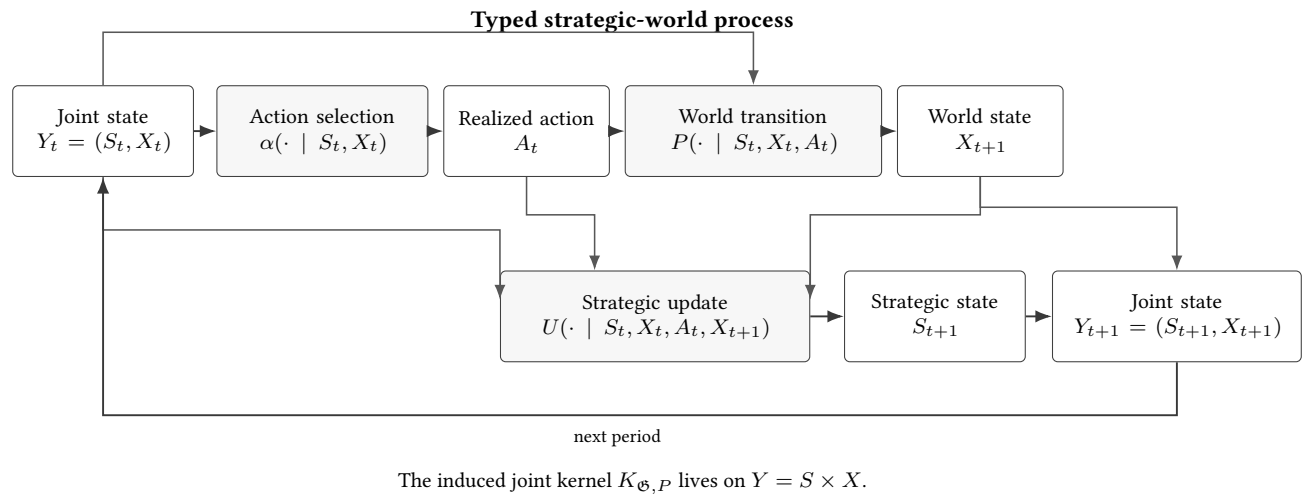


Figure 1: Canonical typed strategic-world process. The joint state $Y_t = (S_t, X_t)$ feeds the action-selection kernel; both the joint state and realized action enter the world transition; and Y_t , A_t , and X_{t+1} enter the strategic update before S_{t+1} and X_{t+1} recombine as Y_{t+1} . The enlarged joint process is ordinary Markov dynamics; the added structure is the declared strategic/world factorization and the intervention semantics.

1.3 Conservative extension of EGR

The set-theoretic relationship requires the intervention grammar to be typed as well as the baseline object. Let PR_ι denote generalized PR status relative to the transported Paper-I protocol class introduced in Section 6.3. Recovery of Paper-I objects is assessed on the embedded EGR class $\iota(\text{EGR})$. Then:

$$\iota(\text{EGR}) \subseteq \text{GR}, \quad \text{PR}_{\text{general}} \subseteq \text{GR}, \quad \text{PR}_\iota \subseteq \text{PR}_{\text{general}}, \quad \iota(\text{PR}_{\text{old}}) = \iota(\text{EGR}) \cap \text{PR}_\iota. \quad (2)$$

Here ι denotes the canonical embedding defined in Section 6. The restriction to PR_ι is essential. The unrestricted overlap $\iota(\text{EGR}) \cap \text{PR}_{\text{general}}$ can be larger because the generalized theory permits constitutive interventions on strategic-update or

auxiliary-state components that have no Paper-I policy counterpart. Thus Paper II conservatively recovers Paper I under transported Paper-I protocols without claiming that every generalized constitutive property of an embedded representation existed in Paper I. The framework retains the distinction between the broad $\text{PR}_{\text{general}}$ class and grounded confirmatory PR status: an auxiliary coordinate can support a confirmatory constitutive claim only when a predeclared relevance map explicitly includes it. The confirmatory GR layer retains basin-reachable descriptor nondegeneracy and separately reports whether descriptor richness remains forward-active after time zero. Conversely, a non-equilibrium learning process can be a generalized PR if a component of its strategic generator is constitutive of the declared regime property. The application-certificate layer leaves these set-theoretic relations unchanged: empirical provenance, identification, support, or sampling uncertainty may downgrade an application certificate without changing the underlying formal GR/PR class.

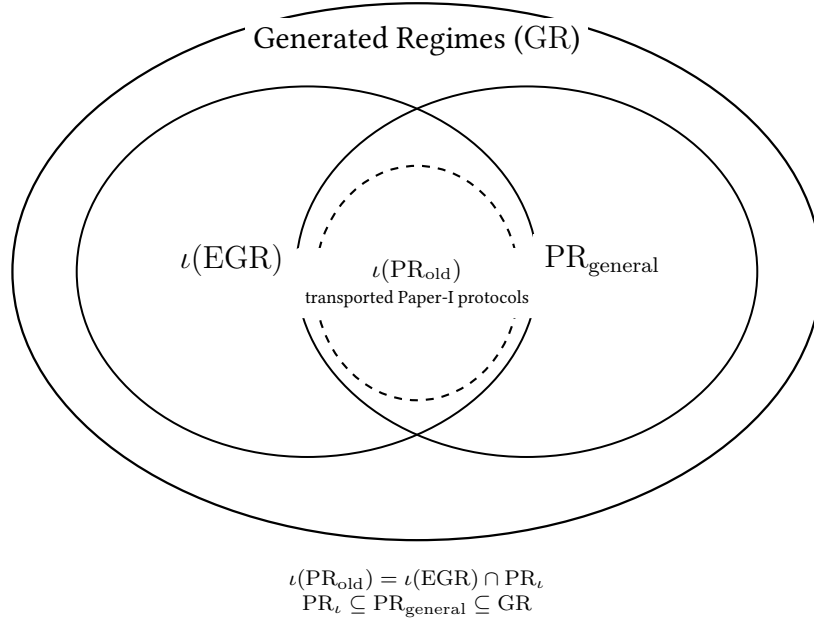


Figure 2: Class hierarchy. Transported Paper-I protocol status is denoted PR_l ; exact recovery of Paper-I Permansson regimes is the dashed intersection $\iota(\text{PR}_{\text{old}})$. The unrestricted $\iota(\text{EGR})/\text{PR}_{\text{general}}$ overlap may be larger. The grounded confirmatory subclass $\text{PR}_g \subseteq \text{PR}_{\text{general}}$ is omitted for readability.

1.4 Scoped novelty claim and nonclaims

The novelty claim is organizational, diagnostic, and semantic rather than a priority claim over the underlying mathematics. Benaïm and Hirsch (1996) provide a unified topological framework for limit sets of asymptotic pseudotrajectories, including fictitious play. Benaïm, Hofbauer, and Sorin (2005) extend this approach to differential inclusions. Sandholm (2015) explicitly separates population games from revision protocols and studies the induced evolutionary dynamics. Weitz et al. (2016), Tilman, Plotkin, and Akçay (2020), and Ito and Yamamichi (2024) couple strategic behavior to environmental states and classify fixed points, bistability, cycles, and related outcomes. Mertikopoulos, Hsieh, and Cevher (2024) develop a common stochastic-approximation template for many learning algorithms. Akin (2009), Kloeden and Rasmussen (2011), and the broader dynamical-systems literature already supply general asymptotic languages. Aubin (1990) supplies viability and controlled-invariance machinery. Bhatt and Borkar (1996) characterize occupation measures in controlled Markov processes.

Accordingly, this paper does not claim: (i) new mathematics of attractors, recurrence, invariant measures, chain recurrence, or viability; (ii) a new equilibrium solution concept; (iii) that strategic/world factorizations are identified from trajectories alone; (iv) that every persistent trajectory is a GR; (v) that finite-horizon persistence is equivalent to exact persistence; or (vi) that a signed feedback diagram by itself provides causal identification. The specific proposed object is a portable typed regime layer with ex ante classification and intervention-relative constitutive semantics.

2. Related Literature and the Remaining Distinction

2.1 Population and evolutionary games

Population-game theory already contains one of the most important abstractions for the present paper. Sandholm (2015) models large populations using revision protocols that generate stochastic evolutionary processes and deterministic mean dynamics. Fixing a revision protocol yields a map from population games to differential equations, and different revision principles can induce qualitatively different dynamics. This is a direct predecessor of the strategic-generator idea. The distinction retained here is that the generated behavior is subsequently composed with an explicitly typed world-transition mechanism and then classified by a regime specification that remains unchanged across admissible generator classes.

Arcak and Martins (2021) offer another close architectural analogue by separating a payoff dynamics model from an evolutionary dynamics model and studying stability compositionally. Their target is convergence to Nash-like rest points under dissipativity conditions. The present framework instead treats convergence, cycles, invariant occupation laws, or other declared regime forms as possible outputs, and it preserves a separate constitutive intervention grammar.

2.2 Learning and stochastic approximation

Learning in games provides a large family of non-equilibrium generators. Mertikopoulos, Hsieh, and Cevher (2024) develop a unified stochastic-approximation template spanning gradient methods, multiplicative weights, optimistic methods, bandit methods, and related algorithms. Benaïm and Hirsch (1996) and Benaïm, Hofbauer, and Sorin (2005) show how broad adaptive processes can be analyzed through internally chain-recurrent or internally chain-transitive limit sets. Galla and Farmer (2013) show that reinforcement-learning dynamics in complicated games can exhibit fixed points, multiplicity, limit cycles, and high-dimensional chaos. Milionis et al. (2023) prove an impossibility result showing that Nash convergence cannot be a universal requirement for game dynamics.

These results strongly motivate separating generator type from regime type. A learning generator may produce a fixed point, cycle, chaotic attractor, or invariant distribution depending on the model. “Non-equilibrium” is therefore not itself a regime classification. It is a statement about the generator or the absence of a particular solution certificate.

2.3 Strategy-environment feedback

Feedback-evolving games are the closest substantive game-facing neighbor. Weitz et al. (2016) explicitly model accumulated strategic behavior as changing the commons and thereby changing subsequent payoffs. Tilman, Plotkin, and Akçay (2020) generalize the framework by separating intrinsic environmental dynamics, strategic impact on the environment, and strategy-update dynamics. Ito and Yamamichi (2024) provide a broad qualitative classification for feedback-evolving games, including stable equilibria, bistability, transient changes in game structure, and persistent oscillation.

These models already establish the conceptual legitimacy of strategic behavior $\leftrightarrow$ world feedback and of regime-like qualitative classification. The remaining distinction is portability: the canonical feedback-evolving model is usually tied to a specific evolutionary update equation or family. The present paper instead treats the strategic generator as a typed replaceable input while keeping the regime specification and constitutive semantics fixed.

2.4 Dynamical systems and the enlarged-state objection

The strongest reduction argument against the present framework is mathematically correct: set $Y_t = (S_t, X_t)$, forget the interpretation of the factors, and analyze the induced process with ordinary dynamical-systems or Markov-process tools. Akin’s survey develops invariant sets, attractors, chain recurrence, Lyapunov functions, minimality, and recurrence. Benaïm and Hirsch characterize limit sets of broad nonautonomous pseudotrajectories. Kloeden and Rasmussen develop nonautonomous processes, pullback attractors, switching systems, and random dynamical systems. Conley-theoretic reasoning also enters modern game dynamics, as emphasized by Milionis et al. (2023).

This paper accepts that reduction at the level of baseline dynamics. Its claim is that the reduction loses information required for constitutive questions. If two different strategic/world factorizations induce the same K , they are dynamically indistinguishable at baseline. Yet an intervention on the strategic generator may leave one K unchanged and alter the other. The factorization is therefore irrelevant to baseline dynamical classification but relevant to typed counterfactual constitution. Sections 6 and 7 formalize this distinction.

2.5 Viability, controlled Markov processes, and institutions

Viability theory provides mature mathematics for remaining inside declared sets under control and for computing viability kernels and capture basins (Aubin, 1990). Controlled Markov process theory similarly provides policy-induced occupation measures and long-run state-action structure (Bhatt and Borkar, 1996). These are important mathematical substrates for particular GR descriptors and persistence criteria rather than competing vocabularies that must be replaced.

Institutional theory supplies a conceptual analogue at a different level. Greif and Laitin (2004) ask how self-enforcing institutions persist in changing environments and how processes unleashed by institutions can reinforce or undermine them. Their quasi-parameter and self-reinforcement concepts are closely related to the idea that behavior changes a world state that feeds back into future strategic conditions. The present framework aims to provide a more abstract process-and-intervention language that can host such mechanisms without identifying institutions with equilibrium by definition.

2.6 Causal abstraction and causal games

Causal abstraction provides the closest existing formal language for the intervention-sensitive equivalence problem underlying Sections 7.2–7.4. Rubenstein et al. (2017) introduce exact transformations between structural equation models and explicitly require agreement about the effects of mapped interventions across levels of description. Beckers and Halpern (2019) refine this idea through increasingly restrictive notions of causal abstraction; Otsuka and Saigo (2022) give a category-theoretic equivalence criterion under which intervention calculi translate consistently; and Geiger et al.

(2025) generalize causal abstraction from mechanism replacement to broader mechanism transformations. These results already establish that representation-level or baseline agreement is not by itself the relevant notion of equivalence when interventions matter.

Causal games bring that same issue directly into strategic settings. Hammond et al. (2023) define structural causal games with mechanised representations of decision rules and provide prediction, intervention, and counterfactual semantics. Mishra, Fox, and Wooldridge (2024) further characterize primitive interventions in causal games, including distinctions that determine whether agents may adapt their policies after an intervention. These literatures therefore substantially anticipate the general intervention-sensitive principle used here. What remains distinct is not intervention semantics itself, but the Generated-Regime object: a reusable ex ante persistent-regime specification that can be applied across heterogeneous strategic generators, together with the Permansson subclass whose membership is defined by constitutive strategic intervention response. The present framework should therefore be read as a regime-classification layer compatible with causal abstraction and causal-game machinery, not as a replacement for either.

2.7 Labelled Markov processes, killed processes, and perturbation theory

Labelled Markov-process theory is the closest process-theoretic prior art for the intervention-family layer. Desharnais et al. (2004) construct behavioral metrics whose zero set is bisimilarity, while Danos et al. (2006) formulate event bisimulation through sub- σ -algebras on general measurable spaces. These results already establish that families of transition kernels indexed by labels admit exact and approximate behavioral equivalence notions. The contribution here is therefore not the generic idea of a label-indexed kernel family; it is the sufficient bridge from a frozen family of typed strategic interventions to GR/PR regime and constitution statements.

Killed-process theory supplies the exact finite-survival and quasi-stationary machinery used in Section 4. Benaïm, Cloez, and Panloup (2018) analyze quasi-stationary distributions for killed Markov processes on compact spaces, while Coolen-Schrijner and van Doorn (2006) and van Doorn and Pollett (2009) demonstrate why QSD existence, uniqueness, and limiting conditional behavior require separate hypotheses. The present paper uses only the elementary sub-Markov identities universally; QSD claims are optional certified subclasses.

Finally, the quantitative layer draws on established perturbation theory rather than presenting a universal new sensitivity theorem. Mitrophanov (2005) derives perturbation bounds for uniformly ergodic Markov chains, and Rudolf and Wang (2025) quantify stability of quasi-stationary laws for killed processes under generator perturbations. Such results enter only as subclass inputs; the general theorem here is the metric margin bridge from bounded property-output error to preserved constitutive classification.

Table 1: Closest neighboring literatures. The framework deliberately uses established mathematical objects where possible; the proposed addition is the typed regime and intervention layer.

Literature	Generator / process	Persistent object	What remains distinct here
Population games	Revision protocols	Rest points, convergence, cycles, stochastic stability	Separate world kernel and portable regime/intervention semantics
Learning in games	Broad learning algorithms	Attractors, limit sets, invariant measures	Same GR definition across generator classes
Feedback-evolving games	Usually replicator / evolutionary updates	Fixed points, bistability, cycles	Generator replacement plus typed constitution
Dynamical systems	Arbitrary law on enlarged state	Invariant sets, recurrence, attractors	Strategic/world semantics and intervention type
Causal abstraction / causal games	Structural mechanisms, decision rules, typed interventions	Intervention-preserving equivalence; causal/counterfactual predictions	Portable persistent-regime object and PR membership rule
Labelled Markov / bisimulation	Label-indexed transition kernels	Behavioral equivalence and metrics	Typed intervention-family signature plus GR/PR semantic preservation
Killed / quasi-stationary processes	Sub-Markov kernels and absorption	Survival, exit laws, QSDs	Exact finite-persistence calculus with optional QSD certification of quasi-regimes
Markov perturbation theory	Baseline/perturbed kernels	Sensitivity of finite or long-run laws under hypotheses	Property-output error bridge to constitutive classification
Controlled Markov / viability	Policies or controls	Occupation measures, viable sets, capture basins	Game-facing regime property and constitutive decomposition
Endogenous institutions	Case-specific repeated-game mechanisms	Persistence, reinforcement, endogenous change	Portable process-level formalization

3. Canonical Strategic-World Environment

3.1 Measurable spaces and states

The canonical paper uses discrete time $t \in \mathbb{N}_0$ and a Markovian representation. Continuous-time, differential-inclusion, and nonautonomous extensions are discussed later. Let $(S, B(S))$ and $(X, B(X))$ be nonempty Polish spaces representing the strategic state and the world state. Let $Y = S \times X$ with the product Borel σ -algebra. Where a measure-based non-triviality

test is used, fix a nonzero Borel reference measure m on Y as part of the model before any regime region is declared. The strategic state may include policies, learning weights, beliefs, memory variables, population strategy frequencies, internal organizational rules, or other variables required to generate future behavior. The world state contains the material, institutional, environmental, network, informational, or other state altered through behavior.

The decomposition is analytical rather than ontological. If a variable is required to make the mechanism Markovian, it must be included in S or X . If a distinction cannot be empirically identified, the model may still declare it structurally, but Section 7 shows that the factorization is not generally identified from the induced joint process.

3.2 Actions and feasibility

Let $I = \{1, \dots, n\}$ be a finite player set when a multi-player interpretation is available. For each player i , let A_i be a nonempty standard Borel action space and let $A_i(y) \subseteq A_i$ be a nonempty feasible-action correspondence with measurable graph. Let $A = \prod_i A_i$ and $A(y) = \prod_i A_i(y)$. For a single-agent, population-level, or exogenous formulation, take A directly to be any nonempty standard Borel behavioral-output space with a nonempty measurable feasible correspondence $A(y)$.

3.3 The strategic generator

The strategic generator is the ordered pair

$$\mathfrak{G} = (\alpha, U). \quad (3)$$

The action-selection kernel $\alpha(da \mid s, x)$ is a Markov kernel from Y to A satisfying feasibility:

$$\alpha(A(s, x) \mid s, x) = 1 \quad \text{for every } (s, x) \in Y. \quad (4)$$

The strategic-update kernel $U(ds' \mid s, x, a, x')$ is a Markov kernel from $S \times X \times A \times X$ to S . It updates the internal strategic state after the action and the realized next world state are available. This ordering is intentionally more general than a single kernel $\Gamma(da, ds' \mid s, x)$: learning rules often update beliefs, propensities, or policies only after observing consequences, rewards, or public signals. Any additional observation can be included explicitly or encoded in the augmented next state.

3.4 World transition

Let $P(dx' \mid s, x, a)$ be a Markov kernel from $Y \times A$ to X . Only its values on feasible state-action pairs matter because α is supported on $A(s, x)$. If an application initially specifies a kernel only on the Borel feasible graph, fix a measurable extension off that graph before applying the canonical formulas; because X is nonempty, a fixed probability law on X can be used off the feasible graph. P contains both intrinsic world dynamics and action-dependent consequences. In a game-environment model, for example, P may encode resource renewal together with depletion caused by current behavior. In an institutional model it may encode how actions alter operative rules, status, capacity, or other continuation-relevant state.

3.5 Induced joint kernel

For a measurable rectangle $C \times D \subseteq S \times X$ define

$$K_{\mathfrak{G}, P}(C \times D \mid s, x) = \int_A \int_D U(C \mid s, x, a, x') P(dx' \mid s, x, a) \alpha(da \mid s, x). \quad (5)$$

Equivalently, for any $E \in B(Y)$,

$$K_{\mathfrak{G}, P}(E \mid s, x) = \int_A \int_X \int_S \mathbf{1}_E(s', x') U(ds' \mid s, x, a, x') P(dx' \mid s, x, a) \alpha(da \mid s, x). \quad (6)$$

Theorem 3.1 (Joint-process well-posedness). Under the measurability and feasibility assumptions above, $K_{\mathfrak{G}, P}$ is a Markov kernel on Y . For every initial law $\lambda_0 \in \Delta(Y)$, there exists a unique probability law $P_{\lambda_0}^{\mathfrak{G}, P}$ on $\Omega = Y^{\mathbb{N}_0}$ whose canonical coordinate process has initial law λ_0 and transition kernel $K_{\mathfrak{G}, P}$.

Proof. Kernel measurability follows from standard composition of measurable kernels: first integrate the bounded measurable indicator $\mathbf{1}_E(s', x')$ against U , then P , then α . The resulting map $y \mapsto K(E \mid y)$ is measurable for every Borel E , and $K(Y \mid y) = 1$. Since S and X are Polish, $Y = S \times X$ is Polish and hence standard Borel. Ionescu-Tulcea then yields a unique probability measure on the countable product path space with the stated initial law and transition kernel. Q.E.D.

Proposition 3.2 (Enlarged-state reduction). Every canonical strategic-world model is an ordinary Markov process on the enlarged state $Y = S \times X$. Consequently, every baseline property that depends only on the induced path law can, in principle, be studied without retaining the strategic/world factorization.

Proof. Immediate from Theorem 3.1. The proposition is a reduction, not a novelty claim. Its importance is diagnostic: any claimed contribution of the typed factorization must concern semantics or interventions not recoverable from K alone. Q.E.D.

Table 2: Core notation.

Symbol	Meaning
S_t	Strategic state
X_t	World state
$Y_t = (S_t, X_t)$	Joint state
A_t	Realized strategic behavior
α	Action-selection kernel
U	Strategic-update kernel
$\mathfrak{G} = (\alpha, U)$	Strategic generator
P	World-transition kernel
$K_{\mathfrak{G}, P}$	Induced joint kernel on Y
$P_{\lambda}^{\mathfrak{G}, P}$	Induced path law from initial law λ
$\Delta(E)$	Borel probability measures on measurable space E
$f_{\#}\mu$	Pushforward of μ under measurable map f
δ_y	Dirac probability law at y
B	Predeclared regime region
B_0	Predeclared initial basin
B_1	Predeclared constitutive comparison set
h	Regime descriptor
ν	Limiting occupation law
$\mathfrak{c}$	Declared convergence mode
g	Frozen relevance map for grounded confirmatory use
ψ	Defining regime-property map
d_{ψ}	Metric on the property space Z_{ψ}
J	Typed admissible intervention
m	Reference measure for non-triviality test
K_B	Killed/sub-Markov kernel obtained by restricting K to transitions remaining in B
$\Delta_{\psi, J}(y)$	Pointwise constitutive effect distance under intervention J
$\kappa_{\psi, J}(B_1)$	Uniform constitutive margin: infimum of $\Delta_{\psi, J}$ over B_1
$\text{Sig}_{\mathcal{J}}(M)$	Frozen indexed baseline-plus-intervention kernel signature for family $\mathcal{J}$

4. Generated Regimes

4.1 Ex ante regime specification

Let H be a Polish descriptor space. Fix a bounded compatible metric d_H on H —for example, replace any compatible complete metric ρ by $\min\{1, \rho\}$ —and let d_{BL} be the corresponding bounded-Lipschitz/Fortet–Mourier metric on $\Delta(H)$. On a Polish space, d_{BL} metrizes weak convergence of probability measures (Dudley, 2002). A regime specification is

$$\Sigma = (B, B_0, h, \nu, \mathfrak{c}). \quad (7)$$

where $B_0 \subseteq B \subseteq Y$ are Borel sets, $h : Y \rightarrow H$ is Borel measurable, $\nu \in \Delta(H)$, and $\mathfrak{c}$ records the declared convergence mode. The descriptor is defined on all of Y rather than only on B so that the same map remains evaluable under interventions that may force exit from the baseline region.

Proposition 4.1a (Polish descriptor-space generality). Replacing compact metrizability of H by Polishness leaves every definition in Sections 4–7 well posed. In particular, a bounded compatible metric exists, $\Delta(H)$ is metrizable by d_{BL} for weak convergence, and the occupation-law conditions below remain meaningful. Polishness alone does not imply uniform tightness of an arbitrary sequence of occupation laws and does not imply existence of a limiting law.

Proof. Let ρ be any compatible complete separable metric on H and set $d_H = \min\{1, \rho\}$. This bounded metric generates the same topology. The associated bounded-Lipschitz metric metrizes weak convergence on probability measures over a Polish space. The remaining definitions use only Borel measurability of h , probability laws on H , and the declared convergence mode. No compactness step is invoked. A sequence of probability measures on a noncompact Polish space need not be uniformly tight, so existence of ν remains an Exact-GR requirement rather than a consequence of the ambient topology. Q.E.D.

Define the admissible initial-law class

$$\mathcal{D}(B_0) = \{\lambda \in \Delta(Y) : \lambda(B_0) = 1\}. \quad (8)$$

The specification is analyst-relative and is not uniquely determined by the process. To block post hoc regime drawing, the following ex ante non-triviality conditions are imposed. For confirmatory certification, a stronger process-relevant descriptor check is added immediately afterward.

Assumption 4.1 (Ex ante non-triviality). (i) B_0 and B are Borel with nonempty $B_0 \subseteq B$. (ii) The regime region B has nonempty interior or positive measure under the reference measure m fixed with the model before B is declared. (iii) $h(B)$ contains at least two points. (iv) B_0 contains at least two states that induce distinct baseline path laws. (v) B , B_0 , h , ν , and c are declared before the evaluated trajectory is classified. The same h is used in baseline and intervention comparisons.

Because a point-initialized path law on $\Omega = Y^{\mathbb{N}_0}$ includes the deterministic coordinate $Y_0 = y$, condition (iv) is literally equivalent to requiring at least two distinct initial states; it blocks singleton basins but does not by itself guarantee different continuation dynamics. Shifted continuation-law diversity $P_y \circ \theta^{-1} \neq P_{y'} \circ \theta^{-1}$, where $\theta(y_0, y_1, \dots) = (y_1, y_2, \dots)$, remains a useful strengthening diagnostic. For grounded confirmatory work, an additional optional response-profile diagnostic may compare the g -projected baseline and frozen-intervention path laws: two states can be substantively distinguishable if their grounded law differs at baseline or under the declared intervention. Neither baseline grounded diversity nor shifted-law diversity is a universal theorem precondition, because latent types, beliefs, or learning states may pool at baseline while responding differently to the counterfactual, and contraction-to-attractor regimes may lose forward diversity immediately. Condition (v) does not prohibit exploratory model development, but exploratory regions must be frozen and evaluated on held-out simulations, data, or cases before confirmatory classification. For confirmatory certification, define the basin-reachable descriptor-law family

$$\mathcal{M}_h(B_0) = \{h_{\#}(\delta_y K_{\mathfrak{G},P}^t) : y \in B_0, t \in \mathbb{N}_0\}.$$

A specification is called descriptor-nondegenerate when $\mathcal{M}_h(B_0)$ contains at least two distinct probability measures. This blocks an unreachable decorative state from supplying all descriptor variation. A stronger diagnostic is forward descriptor richness: let $R_h^+(B_0)$ be the closure in H of the union of $\text{supp}(h_{\#}(\delta_y K_{\mathfrak{G},P}^t))$ over $y \in B_0$ and integers $t \geq 1$; the specification is forward-descriptor-rich when $R_h^+(B_0)$ contains at least two points. Forward richness is not a universal certification gate. A one-step contraction from a nontrivial basin to a common fixed point can be a perfectly legitimate Exact GR. Confirmatory reports should therefore state whether nondegeneracy is forward-active or basin-only; basin-only status is not a failure, but it should not be described as persistent descriptor variation.

4.2 Exact and finite-horizon persistence

Define the first exit time

$$\tau_B = \inf\{t \geq 0 : Y_t \notin B\}. \quad (9)$$

The regime region is exactly invariant under $K_{\mathfrak{G},P}$ when

$$K_{\mathfrak{G},P}(B \mid y) = 1 \quad \text{for every } y \in B. \quad (10)$$

Proposition 4.2 (Equivalent forms of exact invariance). For a Markov process with kernel $K_{\mathfrak{G},P}$, condition (10) is equivalent to $P_y^{\mathfrak{G},P}(\tau_B = \infty) = 1$ for every $y \in B$.

Proof. If $K(B \mid y) = 1$ on B , induction gives $P_y(Y_t \in B \text{ for all } t \leq T) = 1$ for every finite T . Countable intersection over T yields indefinite retention almost surely. Conversely, if indefinite retention holds from every $y \in B$, then in particular $P_y(Y_1 \in B) = K(B \mid y) = 1$. Q.E.D.

For $L \in \mathbb{N}$ and $\eta \in [0, 1]$, define (L, η) -persistence by

$$\inf_{y \in B} P_y^{\mathfrak{G},P}(\tau_B > L) \geq 1 - \eta. \quad (11)$$

For measurable B define the killed (sub-Markov) kernel on B by

$$K_B(y, C) = K_{\mathfrak{G},P}(C \cap B \mid y), \quad y \in B, C \in \mathcal{B}(Y). \quad (11a)$$

The killed kernel records exactly the mass that remains in B after each transition; missing mass is the probability already absorbed at the cemetery state.

Theorem 4.2a (Exact killed-kernel persistence calculus). For every $y \in B$ and integer $n \geq 0$,

$$P_y^{\mathfrak{G},P}(\tau_B > n) = (K_B^n \mathbf{1}_B)(y). \quad (11b)$$

Moreover, with values in $[0, \infty]$,

$$E_y^{\mathfrak{G},P}[\tau_B] = \sum_{n=0}^{\infty} (K_B^n \mathbf{1}_B)(y). \quad (11c)$$

Proof. Because $y \in B$, the event $\{\tau_B > n\}$ is exactly $\{Y_1 \in B, \dots, Y_n \in B\}$. For $n = 0$ both sides equal one. If the identity holds at n , the Markov property gives

$$P_y(\tau_B > n+1) = \int_B P_z(\tau_B > n) K_{\mathfrak{G},P}(dz | y) = (K_B^{n+1} \mathbf{1}_B)(y),$$

proving (11b) by induction. Equation (11c) is the tail-sum identity for a nonnegative integer-valued random variable, with monotone convergence allowing the value $+\infty$. Q.E.D.

Corollary 4.2b (Exact finite-persistence characterization). Condition (11) is equivalent to

$$\inf_{y \in B} (K_B^L \mathbf{1}_B)(y) \geq 1 - \eta. \quad (11d)$$

Corollary 4.2c (One-step lower bound). If $K_{\mathfrak{G},P}(B | y) \geq q$ for every $y \in B$, then $(K_B^L \mathbf{1}_B)(y) \geq q^L$ for every $y \in B$. This is a lower bound on the exact quantity (11d), not a replacement for it, and when $q < 1$ it supplies no positive lower bound on indefinite survival.

Finite-horizon survival remains strictly weaker than exact persistence. Equations (11b)–(11d) identify the exact finite-horizon object; Proposition 4.2 continues to characterize the separate exact-invariance boundary. The labels exact GR, quasi-regime, and metastable extension are therefore kept distinct.

4.3 Descriptors and limiting occupation laws

For the descriptor process $h(Y_t)$, define the empirical occupation measure

$$\hat{\nu}_T^h = \frac{1}{T} \sum_{t=0}^{T-1} \delta_{h(Y_t)}. \quad (12)$$

The canonical convergence modes are:

- Almost sure weak convergence: $d_{\text{BL}}(\hat{\nu}_T^h, \nu) \rightarrow 0$ almost surely for every $\lambda_0 \in \mathcal{D}(B_0)$.
- Convergence in probability under d_{BL} : for every $\varepsilon > 0$, $P(d_{\text{BL}}(\hat{\nu}_T^h, \nu) > \varepsilon) \rightarrow 0$ for every admissible λ_0 .
- Mean bounded-Lipschitz convergence: $E[d_{\text{BL}}(\hat{\nu}_T^h, \nu)] \rightarrow 0$ for every admissible λ_0 .
- Convergence in distribution of the random occupation measure: $\hat{\nu}_T^h \Rightarrow \nu$ as random elements of $\Delta(H)$, equivalently $\mathcal{L}(\hat{\nu}_T^h) \Rightarrow \delta_\nu$. Because the target ν is deterministic and $\Delta(H)$ is metrizable, this mode is equivalent to convergence in probability to ν ; it is listed explicitly to preserve the Paper-I convergence vocabulary.
- These modes are canonical rather than exhaustive: $\mathfrak{c}$ may specify another precisely defined convergence mode, provided the mode is fixed ex ante and is meaningful for every admissible initial law.

A limiting occupation law is not automatically an invariant law of a Markov factor. Call h a Markov factor for $K_{\mathfrak{G},P}$ on B if there exists a Markov kernel K_h on H such that $K_{\mathfrak{G},P}(h^{-1}(C) | y) = K_h(C | h(y))$ for every $y \in B$ and every Borel $C \subseteq H$. Only when such a factor kernel is available and $\nu K_h = \nu$ is ν called invariant on the factor space; otherwise ν remains an occupation-law descriptor of the full process.

Definition 4.3 (Exact Generated Regime). An Exact Generated Regime is a tuple $R = (\mathfrak{G}, P, B, B_0, h, \nu, \mathfrak{c})$ such that: (i) the canonical process is well posed; (ii) Assumption 4.1 holds; (iii) B is exactly invariant under $K_{\mathfrak{G},P}$; and (iv) ν is a limiting occupation law for every $\lambda_0 \in \mathcal{D}(B_0)$ under the declared convergence mode $\mathfrak{c}$. An Exact GR is called descriptor-nondegenerate when, in addition, the basin-reachable descriptor-law family $\mathcal{M}_h(B_0)$ defined above contains at least two distinct measures. Confirmatory GR certification in this paper uses the descriptor-nondegenerate subclass; the broader Exact-GR definition is retained for exact backward compatibility with Paper I. Forward-descriptor-rich status is a separately reported stronger diagnostic, not an additional universal gate.

Common admissible descriptors include fixed-point or periodic factor dynamics, recurrence classes, occupation shares, balanced growth after normalization, contraction, approach to a target set, or a finite symbolic regime code. Labels such as “stable,” “chaotic,” “self-reinforcing,” or “transitioning” must be tied to explicit topological, probabilistic, or intervention-relative conditions rather than used as free interpretive adjectives.

4.4 Quasi-regimes and metastable extensions

An (L, η) -quasi-regime freezes the same ex ante region/basin/descriptor specification and satisfies the finite-horizon survival bound (11), but it is not an Exact GR and need not possess a limiting occupation law unless that is established separately. A metastable GR extension additionally declares a disturbance or approximation family $\{K_\varepsilon\}_{\varepsilon>0}$, a region B , a basin B_0 , and an integer-valued exit-time scale $r : (0, \varepsilon_0] \rightarrow \mathbb{N}$ with $r(\varepsilon) \rightarrow \infty$ as $\varepsilon \downarrow 0$. A minimal metastability requirement is

$$\lim_{\varepsilon \downarrow 0} \inf_{y \in B_0} P_y^{K_\varepsilon}(\tau_B > r(\varepsilon)) = 1. \quad (13)$$

This definition is intentionally weak and is not presented as a universal metastability concept. An optional stronger subclass based on killed-process theory is defined next, while remaining separate from Exact GR.

Definition 4.4a (QSD-certified quasi-regime). An (L, η) -quasi-regime is QSD-certified by a pair (μ_B, θ_B) if, in addition to satisfying the finite-persistence gate (11), its killed kernel K_B admits a probability measure μ_B on B and a survival factor $\theta_B \in (0, 1]$ such that

$$\mu_B K_B = \theta_B \mu_B. \quad (13a)$$

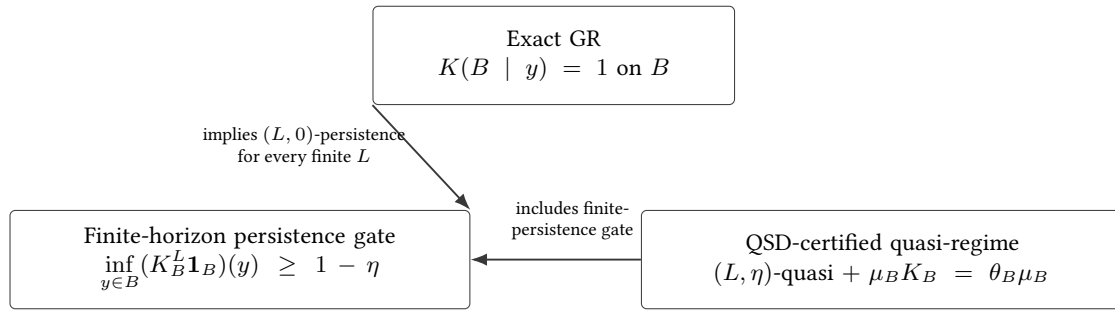
The eigenmeasure relation (13a) is an additional certificate for the killed dynamics; by itself it does not imply the uniform statewise finite-persistence requirement in (11).

Proposition 4.4b (QSD survival and conditional stationarity). If (13a) holds, then for every $n \geq 0$,

$$P_{\mu_B}(\tau_B > n) = \theta_B^n, \quad \mathcal{L}(Y_n \mid \tau_B > n) = \mu_B. \quad (13b)$$

Proof. Iterating (13a) gives $\mu_B K_B^n = \theta_B^n \mu_B$. Evaluating on B yields survival probability θ_B^n by Theorem 4.2a. Normalizing the subprobability $\mu_B K_B^n$ by that survival mass returns μ_B , proving conditional stationarity. Q.E.D.

Existence and uniqueness are additional hypotheses, not consequences of the GR definition. Discrete-time killed chains can have zero, one, or infinitely many QSDs when killing occurs from infinitely many states (Coolen-Schrijner and van Doorn, 2006), and reducible finite-state chains require explicit conditions for uniqueness and for identification of the limiting conditional law (van Doorn and Pollett, 2009). A QSD need not coincide with a Yaglom limit or a quasi-ergodic law. Accordingly, the phrase QSD-certified quasi-regime is used only when both the finite-persistence gate and the particular QSD claim have been separately established. Stronger spectral, exponential-exit, or metastability conditions remain application-specific. The eigenmeasure formulation is standard quasi-stationary-distribution structure (Benaïm, Cloez, and Panloup, 2018).



The QSD equation alone does not imply the uniform finite-persistence gate.

Figure 3: Persistence implications and QSD certification. Exact GR implies $(L, 0)$ -persistence at every finite horizon. A QSD-certified quasi-regime separately includes the finite-persistence gate and an eigenmeasure certificate; the QSD equation alone does not establish the uniform gate.

4.5 Regime locus

Because Y is typed, a GR can be classified by locus without changing the underlying mathematics. A world-locus regime has $B = S \times B_X$ and a descriptor $h = h_X \circ \text{pr}_X$. A strategic-locus regime has $B = B_S \times X$ and $h = h_S \circ \text{pr}_S$. A coupled regime uses a genuinely joint region or descriptor. This typing is semantic information that is discarded by the enlarged-state reduction but can matter for interpretation and interventions. The framework retains that semantic declaration operational in confirmatory PR analysis through the relevance map introduced next: a world-locus claim will normally project away representation-only strategic coordinates, whereas a strategic or coupled claim may explicitly retain them.

5. Permansson Regimes and Typed Constitution

5.1 Why constitution is separate from persistence

A process can satisfy a GR specification for reasons that are exogenous to the strategic generator. A favorable resource trend, an absorbing world state, or a structural transition mechanism may create the declared regime even if strategic behavior is irrelevant. Permansson classification therefore adds an intervention-relative test: a strategic component is constitutive only if a fixed admissible intervention on that component changes a predeclared defining regime property while the declared non-intervened primitives are held fixed.

5.2 Regime-property maps

Fix a GR specification Σ . Let Z_ψ be a metric space with metric d_ψ , and let

$$\psi : \Delta(\Omega) \longrightarrow Z_\psi. \quad (14)$$

be a predeclared regime-property map on admissible state-path laws. No measurability assumption on ψ is needed for the pointwise constitutive definitions (15)-(16); when an application treats estimated or random path laws as random elements and then integrates, optimizes, or otherwise randomizes ψ , the corresponding measurability or regularity must be imposed explicitly. The dependence of ψ on the frozen descriptor and specification may be suppressed in notation. Examples include exact persistence status, long-run growth class, limiting composition, occupation mass on a target subset, strategic-mode preservation, or a vector of such properties. For confirmatory PR analysis, also freeze a Borel relevance map $g : Y \rightarrow G$ into a declared standard-Borel regime-observation space G . The map must be rich enough to represent the baseline regime observables: $B = g^{-1}(B_G)$ for some Borel $B_G \subseteq G$ and $h = h_G \circ g$ for some Borel $h_G : G \rightarrow H$. Writing g^∞ for the coordinatewise path map, ψ is g -grounded when there exists a predeclared map ψ_G on path laws over G such that $\psi(\mu) = \psi_G((g^\infty)_\# \mu)$ for every admissible baseline or intervened path law. Thus the constitutive comparison may use only state information explicitly declared by g . If a claimed property depends on realized actions, a sufficient action-mode record must be included in S and in g before the specification is frozen. For a confirmatory PR claim, the semantic counterfactual pipeline is frozen jointly before the counterfactual outcome is evaluated, even though the GR gate may be checked first. The frozen pipeline includes g , ψ , B_1 , the component grammar, the intervention protocol J , and any hard structural target/transport conventions used to define that intervention. If the pipeline is selected from data rather than predeclared, confirmatory use requires a valid held-out or selection-adjusted procedure that covers the search. The broad PR_{general} class remains mathematically available without this extra relevance restriction; grounded status is the confirmatory subclass used to exclude undeclared representation padding.

Confirmatory applications should also conduct a relevance-map audit. The audit should justify the substantive information retained by g ; verify that B , h , and ψ factor through g as claimed; search strict coarsenings when feasible; report non-uniqueness when several incomparable sufficient relevance maps exist; and rerun the classification under defensible alternatives when the semantic choice is not unique. A coordinate is not substantively relevant merely because retaining it increases B_1 diversity or predictive fit. The framework does not assume that one uniquely minimal or canonical relevance map always exists.

5.3 Admissible typed interventions

Interventions are typed by the object they modify.

Formally, fix the ambient canonical environment $(S, X, A, A(\cdot), \alpha, P, U)$. An admissible typed intervention J is a predeclared map that returns replacement kernel(s) on the same declared measurable spaces, satisfies the relevant measurability and feasibility conditions, and states which elements of the environment are held fixed. The label k or ℓ identifies the predeclared target component; constitution is always relative to this intervention protocol rather than to an intrinsic decomposition of a kernel into unnamed parts.

- Action-selection intervention J_k^α : replaces a predeclared component k of α with a feasible measurable rule α' while holding U , P , the feasible-action correspondence, and all other declared primitives fixed.
- Strategic-update intervention J_k^U : replaces a predeclared component k of U with a measurable update kernel U' while holding α and P fixed.
- Strategic-generator intervention $J_k^\mathfrak{G}$: changes a declared component of α , U , or both, with all unaffected components listed explicitly.
- World-transition intervention J_ℓ^P : replaces a declared component of P while holding $\mathfrak{G}$ fixed.
- Institutional or structural intervention: changes feasibility, information, admission rules, or other model primitives. If the original generator becomes infeasible, the protocol must declare a replacement admissible behavior rule. Such an intervention is not silently reclassified as a generator intervention.

Two admissibility levels must be kept separate. Formal admissibility is the kernel-level requirement above: the intervention is well typed, measurable, feasible, and explicit about the objects held fixed. Confirmatory empirical admissibility is stronger. It additionally requires a defensible claim that the held-fixed mechanisms would remain invariant under the intervention over every state/action row that becomes intervention-reachable.

For confirmatory empirical use, fix the predeclared hard structural compatibility relations and let T denote the declared target. Let T^* denote the closure of T under those hard relations. If preserving a hard identity forces another component to move, the intervention must expand/retype its target or be declared inadmissible under that structural specification. Hard identities must be distinguished from soft fitted parameter ties or convenient parametric families; leaving an estimated family is not by itself an admissibility failure.

Causal modularity and support are empirical claim requirements rather than additions to the formal intervention definition. Evidence for a held-fixed mechanism may come from randomized variation, justified sufficient-state adjustment, structural equations, natural experiments, engineering or institutional mechanism knowledge, or another defensible causal design. Full observational action support alone is not sufficient. If an intervention reaches unsupported rows that matter to ψ , or if the stability of the held-fixed mechanism is unresolved, the formal counterfactual may remain mathematically defined while the application records identification status `STRUCTURALLY_UNRESOLVED`; statistical status, if applicable, is reported separately.

The intervened strategic generator need not be an equilibrium or a best response. The purpose of the intervention is diagnostic constitution, not post-intervention equilibrium certification. A date-indexed intervention is admissible in the stationary canonical core only after a clock variable has been included in S ; equivalently it belongs to the nonautonomous extension. Section 6 uses the clock-state construction to embed the full Paper-I intervention grammar.

5.4 Constitutive components

Let $B_1 \subseteq B_0$ be a predeclared Borel set containing at least two states with distinct baseline path laws. Let $J_k^\mathfrak{G}$ be an admissible strategic-generator intervention. A strategic-generator component k is $(\psi, J_k^\mathfrak{G}, B_1)$ -constitutive if

$$\psi(P_y^{\mathfrak{G},P}) \neq \psi(P_y^{J_k^\mathfrak{G}(\mathfrak{G}),P}) \quad \text{for every } y \in B_1. \quad (15)$$

It is uniformly constitutive if there exists $\delta_\psi > 0$ such that

$$\inf_{y \in B_1} d_\psi(\psi(P_y^{\mathfrak{G},P}), \psi(P_y^{J_k^\mathfrak{G}(\mathfrak{G}),P})) \geq \delta_\psi. \quad (16)$$

For quantitative analysis define the pointwise constitutive effect profile and its uniform margin by

$$\Delta_{\psi,J}(y) = d_\psi(\psi(P_y), \psi(P_y^J)), \quad \kappa_{\psi,J}(B_1) = \inf_{y \in B_1} \Delta_{\psi,J}(y). \quad (16a)$$

Thus pointwise constitution requires $\Delta_{\psi,J}(y) > 0$ for every $y \in B_1$, while uniform constitution is equivalent to $\kappa_{\psi,J}(B_1) > 0$. Pointwise constitution alone does not imply a positive uniform margin.

A component ℓ of P is structurally constitutive under an admissible J_ℓ^P when the analogous property holds with $\mathfrak{G}$ fixed. Strategic and structural constitution can coexist. Structural constitution by itself does not make the regime a Permansson regime, preserving the naming logic of the earlier EGR paper.

Definition 5.1 (Generalized Permansson Regime). An Exact GR $R = (\mathfrak{G}, P, B, B_0, h, \nu, \mathfrak{c})$ is a generalized Permansson Regime relative to $(\psi, J_k^\mathfrak{G}, B_1)$ if at least one predeclared component k of the strategic generator is constitutive of a defining regime property ψ under the admissible intervention $J_k^\mathfrak{G}$ on the predeclared set B_1 . Uniform PR status uses condition (16). A generalized PR is grounded relative to a frozen relevance map g when B and h factor through g as specified in Section 5.2 and ψ is g -grounded. The formal grounded subclass used as a prerequisite for confirmatory applications requires both a descriptor-nondegenerate Exact-GR baseline and grounded PR status; denote this subclass PR_g when a compact label is useful. Empirical confirmation additionally requires the application-level certificate of Sections 10.5-10.7.

PR status is therefore relative to a baseline model, regime specification, property map, intervention protocol, and initial-state set; grounded confirmatory status is additionally relative to the frozen relevance map g . It is not an intrinsic label attached to $K_{\mathfrak{G},P}$ alone. The distinction matters under state augmentation: $\text{PR}_{\text{general}}$ deliberately permits any predeclared typed-state property, whereas PR_g refuses to let a coordinate omitted from g create a confirmatory constitutive claim.

Component attribution is grammar-relative. A compound block may be constitutive even when no singleton is; a singleton effect may disappear when components are merged; and several incomparable constitutive blocks may coexist. Confirmatory applications should therefore freeze the component grammar and target before inspection of the counterfactual, record whether the target is atomic, compound, or jointly defined, and test substantively meaningful coarsenings or refinements when attribution matters. Under reparameterization, interventions must be transported by their substantive meaning rather than matched by raw coordinate labels. Constitution under a frozen intervention is a stronger claim than unique attribution to one intrinsically privileged component.

5.4.1 Quantitative perturbation certification

The margin in (16a) permits a general robustness statement that is independent of how the baseline and intervention error bounds were obtained. Let $\tilde{P}_y$ and $\tilde{P}_y^J$ denote perturbed or approximated baseline and intervention path laws evaluated by the same frozen property map ψ .

Proposition 5.2 (Two-sided constitutive-margin perturbation bound). Suppose, uniformly for $y \in B_1$,

$$d_\psi(\psi(P_y), \psi(\tilde{P}_y)) \leq \varepsilon_0, \quad d_\psi(\psi(P_y^J), \psi(\tilde{P}_y^J)) \leq \varepsilon_J. \quad (16b)$$

Then for every $y \in B_1$,

$$|\Delta_{\psi,J}(y) - \tilde{\Delta}_{\psi,J}(y)| \leq \varepsilon_0 + \varepsilon_J, \quad (16c)$$

and, writing $\tilde{\kappa}$ for the perturbed uniform margin on the same B_1 ,

$$|\tilde{\kappa} - \kappa| \leq \varepsilon_0 + \varepsilon_J, \quad \tilde{\kappa} \geq \kappa - \varepsilon_0 - \varepsilon_J. \quad (16d)$$

Proof. For any four points $a, b, \tilde{a}, \tilde{b}$ in a metric space, the triangle inequality and its reverse form imply

$$|d(a, b) - d(\tilde{a}, \tilde{b})| \leq d(a, \tilde{a}) + d(b, \tilde{b}).$$

Apply this with the baseline and intervention property outputs to obtain (16c). If two real-valued functions differ uniformly by at most e , their infima differ by at most e ; applying this with $e = \varepsilon_0 + \varepsilon_J$ yields (16d). Q.E.D.

Corollary 5.3 (Robust PR certificate). If $\kappa_{\psi,J}(B_1) > \varepsilon_0 + \varepsilon_J$, then $\tilde{\kappa} > 0$ and uniform constitutive status survives the perturbation. Equality $\kappa = \varepsilon_0 + \varepsilon_J$ does not certify survival. This theorem is a bridge from property-output error to PR classification; it does not by itself provide ε_0 or ε_J .

Proposition 5.4 (Finite-horizon kernel-to-path envelope). Use the probability-distance convention $\|\mu - \nu\|_{TV} := \sup_A |\mu(A) - \nu(A)|$. Let K and $\tilde{K}$ be Markov kernels on the same state space and let $\varepsilon \in [0, 1]$ satisfy

$$\sup_y \|K(y, \cdot) - \tilde{K}(y, \cdot)\|_{TV} \leq \varepsilon.$$

From the same point initial state y , let $P_y^{0:T}$ and $\tilde{P}_y^{0:T}$ denote the path laws through T transitions. Then

$$\|P_y^{0:T} - \tilde{P}_y^{0:T}\|_{TV} \leq 1 - (1 - \varepsilon)^T \leq T\varepsilon. \quad (16e)$$

Proof. Couple the chains step by step. Conditional on agreement through time t , use a maximal coupling of $K(Y_t, \cdot)$ and $\tilde{K}(Y_t, \cdot)$; the probability of remaining coupled for the next transition is at least $1 - \varepsilon$. Hence the probability of agreement through T transitions is at least $(1 - \varepsilon)^T$. The coupling inequality gives the first bound; Bernoulli's inequality gives the second. Q.E.D.

No infinite-horizon analogue is asserted without additional structure. Small one-step kernel perturbations can cause order-one changes in stationary or infinite-path behavior near loss of ergodicity, and discontinuous ψ can flip under arbitrarily small law perturbations. For long-run occupation laws, invariant laws, or QSDs, kernel-to-law bounds are used only in subclasses satisfying the corresponding uniform-ergodic, contraction, spectral, or killed-process hypotheses (Mitrophanov, 2005; Rudolf and Wang, 2025).

5.5 Protocol scope and negative claims

Definition 5.1 makes PR status relative to the frozen property/intervention protocol (ψ, J, B_1) . Failure of one declared intervention therefore establishes only nonconstitution under that protocol; it is not a global statement that no strategically constitutive intervention exists. For applications, use the label `ROBUST_NONCONSTITUTIVE_UNDER_FROZEN_PROTOCOL` when a declared protocol is excluded over the required state/model set. A broader `NO_PR_IN_DECLARED_INTERVENTION_FAMILY` claim requires exclusion of every member of a predeclared admissible intervention family, or a theorem that rules the family out.

This protocol scope is especially important under partial identification. A negative claim must quantify over the same provenance-valid model set, B_1 , and declared intervention family that define the application. Non-significance, failure to obtain a uniform gap, or failure of one intervention does not by itself establish global non-PR status.

5.6 Regime equivalence

The regime layer supports several deliberately distinct equivalence notions. For the cleanest comparison, let two GRs share a common descriptor space H , descriptor h , convergence mode, and declared comparison set $I \subseteq Y$. Let $h^\infty : \Omega \rightarrow H^{\mathbb{N}_0}$ be the coordinatewise descriptor map $h^\infty((Y_t)_t) = (h(Y_t))_t$.

- Strong descriptor path-law equivalence: $(h^\infty)_\# P_y^{(1)} = (h^\infty)_\# P_y^{(2)}$ for every $y \in I$.
- Occupation equivalence: both satisfy the same declared persistence requirement and convergence mode and have the same limiting occupation law ν on H .
- Property equivalence: $\psi(P_y^{(1)}) = \psi(P_y^{(2)})$ for every y in the declared comparison set on which ψ is evaluated.
- Constitutive equivalence relative to matched interventions J_1, J_2 : baseline property equivalence holds and $\psi(P_y^{(1), J_1}) = \psi(P_y^{(2), J_2})$ for every declared comparison state y ; uniform constitutive equivalence additionally preserves the corresponding d_ψ gaps.

The last notion is strictly stronger than baseline dynamical equivalence. This distinction is the main substantive reason to preserve the strategic/world factorization after conceding the enlarged-state reduction.

6. Embedding the Equilibrium-Generated Theory

6.1 Paper-I EGR object

The earlier EGR core assumes a discounted stochastic game with a Polish payoff-relevant state space X , finite feasible action sets, bounded measurable stage payoffs, a measurable state-transition kernel P , discount factors, and a selected measurable pure stationary Markov-perfect equilibrium π^* . The equilibrium induces $P^{\pi^*}(D | x) = P(D | x, \pi^*(x))$. An EGR is $R_E = (\pi^*, B, B_0, h, \nu, c)$ with ex ante non-triviality, exact invariance of B under P^{π^*} , and the declared occupation-law convergence for every initial law supported on B_0 (Hermansson, 2026a).

6.2 Canonical embedding

For the fully conservative embedding, let A be the finite Paper-I joint action space, adjoin a symbol $\perp \notin A$, and define the strategic state $S_E = \mathbb{N}_0 \times (A \cup \{\perp\})$ with the discrete topology. The coordinate $s_t = (t, a_{t-1})$ records the clock and the previously realized joint action, with $a_{-1} = \perp$. Let

$$\begin{aligned} \alpha^{\pi^*}(da | (t, \bar{a}), x) &= \delta_{\pi^*(x)}(da), \\ U^{\text{rec}}(ds' | (t, \bar{a}), x, a, x') &= \delta_{(t+1, a)}(ds'). \end{aligned} \quad (17)$$

Retain the original world-transition law by setting $P_E(dx' | (t, \bar{a}), x, a) = P(dx' | x, a)$. Let $\iota_0(x) = ((0, \perp), x)$, lift the regime region to $\tilde{B} = S_E \times B$, and lift the initial basin to $\tilde{B}_0 = \iota_0(B_0)$. The action record makes every realized A_t measurable from the next strategic coordinate S_{t+1} ; the clock makes date-indexed Paper-I interventions state-indexed in the enlarged model.

On embedded paths define the measurable decoding map D by $D(((t, \bar{a}_t), x_t)_{t \geq 0}) = ((x_t)_{t \geq 0}, (a_t)_{t \geq 0})$, with $a_t = \bar{a}_{t+1}$. On the null/off-support set of paths that do not satisfy the recording relation, extend D arbitrarily. Thus both the Paper-I world path and the realized action sequence are recoverable from the embedded state path.

$$\begin{aligned} K_{\mathfrak{G}^{\pi^*}, P_E}(S_E \times D | (t, \bar{a}), x) &= P(D | x, \pi^*(x)) \\ &= P^{\pi^*}(D | x). \end{aligned} \quad (18)$$

Because Paper I defines h only on B , choose any $h_0 \in H$ and the Borel extension $\bar{h}(x)=h(x)$ for $x \in B$ and $\bar{h}(x) = h_0$ otherwise; define $\tilde{h}((t, \bar{a}), x) = \bar{h}(x)$. If the measure branch of Paper-I non-triviality is used, lift the reference measure by $\tilde{m} = (\iota_0)_\# m$. If the interior branch is used, $\tilde{B}$ has nonempty interior whenever B does because S_E is discrete. For grounded confirmatory transport, the default relevance map is $\tilde{g}((t, \bar{a}), x) = x$; if a Paper-I property explicitly depends on the recorded action or another transported mode, enlarge $\tilde{g}$ ex ante to include the corresponding sufficient record.

Theorem 6.1 (EGR embedding). The lifted object $\iota(R_E) = (\mathfrak{G}^{\pi^*}, P_E, \tilde{B}, \tilde{B}_0, \tilde{h}, \nu, c)$ is an Exact GR if and only if R_E is an EGR in the earlier theory. Under projection $\text{pr}_X : S_E \times X \rightarrow X$, the baseline world-state path laws, exact persistence status, descriptor process on the baseline region, limiting occupation law, and convergence mode are identical.

Proof. Equation (18) gives the Paper-I equilibrium-induced kernel as the X -marginal of the embedded transition. Starting from any law supported on $\tilde{B}_0$, induction shows that the X -coordinate has exactly the Paper-I path law and that $S_t = (t, A_{t-1})$ records the realized action history. Exact invariance of B implies exact invariance of $\tilde{B}$, and conversely projection of any embedded path in $\tilde{B}$ lies in B . Since $\tilde{h}=h$ on B , the empirical descriptor measures are identical along all baseline paths.

The lifted non-triviality conditions follow from the constructions of $\tilde{B}$, $\tilde{B}_0$ and $\tilde{m}$. Moreover, the basin-reachable descriptor-law family is identical under projection, so descriptor-nondegenerate Paper-I applications remain descriptor-nondegenerate after embedding. The same projection also transports the optional forward-richness diagnostic. Q.E.D.

6.3 Conservative recovery of the original Permansson regime

In Paper I, a policy intervention J_k^π may replace a predeclared policy component on declared states or dates. Let C_I denote the class of admissible Paper-I constitutive property/intervention protocols, and let $\iota_{\#}C_I$ denote the protocols transported into the canonical strategic-world environment by the construction below. Define PR_ι as generalized PR status evaluated only relative to protocols in $\iota_{\#}C_I$; when comparing with Paper I, restrict this status to embedded EGR objects. For $J_k^\pi \in C_I$, let $\rho_t^J(x)$ denote the resulting feasible joint action rule, equal to $\pi^*(x)$ off the intervention set. Map it to the action-selection intervention $\alpha^J(da \mid (t, \bar{a}), x) = \delta_{\rho_t^J(x)}(da)$, while keeping U^{rec} and P_E fixed. Thus date-indexed interventions are represented without changing the canonical Markov form, and the strategic state records the realized intervention action one step later.

Theorem 6.2 (Conservative Permansson extension). Let R_E be an EGR and let (ψ_I, J_k^π, B_1) be an admissible Paper-I constitutive protocol whose property functional is well defined on both the baseline and intervened laws under a frozen descriptor convention. If an intervention may leave B and ψ_I evaluates h after exit, include a predeclared Borel extension $\bar{h}$ in that convention. Define the transported property map on embedded path laws by $\tilde{\psi}(\mu) = \psi_I((\text{pr}_X^\infty)_{\#}\mu, \bar{h})$, or by the corresponding measurable decode $D_{\#}\mu$ when the Paper-I protocol explicitly includes recorded actions. For grounded confirmatory transport, take $\tilde{g}((t, \bar{a}), x) = x$ for world-path properties and enlarge $\tilde{g}$ ex ante to $(x, \bar{a})$, the clock, or another sufficient transported record only when the Paper-I property actually uses that information. Then R_E is a Permansson regime under the Paper-I definition if and only if its embedding $\iota(R_E)$ is a generalized Permansson regime relative to $(\psi, J_k^\pi, \bar{B}_1)$, where $\bar{B}_1 = \iota_0(B_1)$; whenever the transported $\tilde{\psi}$ is $\tilde{g}$ -grounded, grounded status is preserved as well.

Proof. Baseline world-state path laws coincide by Theorem 6.1. Under the mapped intervention, α^J chooses exactly the Paper-I intervened action at each declared date/state, P_E remains the original transition law, and U^{rec} merely records the realized action. Hence the projected state path and, where used, the decoded action path coincide with the Paper-I counterfactual execution. By construction of $\tilde{\psi}$ and the common frozen descriptor convention, baseline and intervention ψ -values are identical to their Paper-I counterparts for every $y \in B_1$, as are the corresponding d_ψ distances. When grounded transport is requested, $\tilde{g}$ contains exactly the state/action record used by the Paper-I property, so $\tilde{\psi}$ factors through $(\tilde{g}^\infty)_{\#}$ by construction. Therefore pointwise, uniform, and grounded constitution are preserved in both directions. Q.E.D.

Theorems 6.1 and 6.2 establish the conservative hierarchy after identifying Paper-I objects with their images under ι : $\iota(\text{EGR}) \subseteq \text{GR}$, $\text{PR}_\iota \subseteq \text{PR}_{\text{general}} \subseteq \text{GR}$, and $\iota(\text{PR}_{\text{old}}) = \iota(\text{EGR}) \cap \text{PR}_\iota$. No equality is asserted with the unrestricted intersection $\iota(\text{EGR}) \cap \text{PR}_{\text{general}}$. The latter may contain additional generalized constitutive classifications that act on strategic-update or auxiliary representation components absent from the Paper-I policy grammar. The grounded confirmatory layer does not alter this broad hierarchy: $\text{PR}_g \subseteq \text{PR}_{\text{general}}$, and transported Paper-I protocols are grounded whenever their relevance map is frozen to the world/action information actually used by the original property.

7. Representation, Identification, and Constitutive Equivalence

7.1 Baseline representation invariance

Proposition 7.1 (Baseline representation invariance). Let two well-posed canonical factorizations $(\mathfrak{G}_1, P_1)$ and $(\mathfrak{G}_2, P_2)$ be defined on the same joint state space Y and satisfy $K_{\mathfrak{G}_1, P_1} = K_{\mathfrak{G}_2, P_2}$. Suppose they use the same frozen regime specification Σ and the same reference measure m whenever the measure branch of Assumption 4.1 is invoked. Then R_1 is an Exact GR if and only if R_2 is an Exact GR, with identical persistence and occupation-law classification.

Proof. The exact-GR definition depends on the baseline model only through the induced kernel and its path laws, together with the common frozen specification. Equality of K gives equality of all finite-dimensional distributions for common initial laws and hence equality of path laws. Exact invariance and descriptor occupation convergence therefore coincide. Q.E.D.

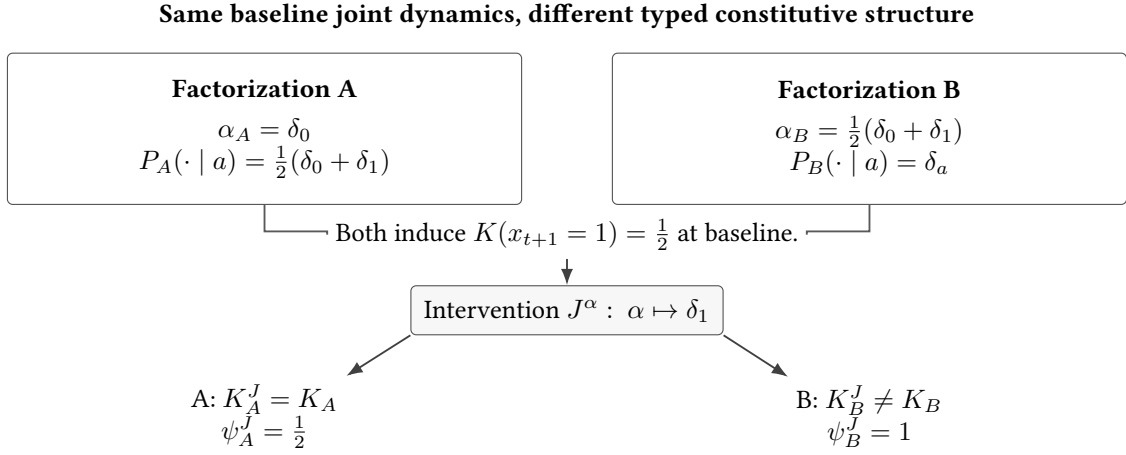
7.2 Factorization is not identified by the joint process

Theorem 7.2 (Factorization non-identification). Equality of induced joint kernels does not imply equality of the strategic generator or world-transition kernel. In general, $K_{\mathfrak{G}_1, P_1} = K_{\mathfrak{G}_2, P_2}$ does not imply $\mathfrak{G}_1 = \mathfrak{G}_2$ or $P_1 = P_2$.

Proof. Consider $S = \{\star\}$, $X = \{0, 1\}$, $A = \{0, 1\}$, and trivial U . In Factorization A, let $\alpha_A(\cdot \mid \star, x) = \delta_0$ and $P_A(\cdot \mid \star, x, a) = (1/2)\delta_0 + (1/2)\delta_1$ for every x, a . In Factorization B, let $\alpha_B(\cdot \mid \star, x) = (1/2)\delta_0 + (1/2)\delta_1$ and $P_B(\cdot \mid \star, x, a) = \delta_a$. Under both factorizations the next world state is Bernoulli(1/2), independent of the current state, so the induced K is identical. Yet $\alpha_A \neq \alpha_B$ and $P_A \neq P_B$. Q.E.D.

The theorem has an immediate epistemic implication: the strategic/world decomposition is a structural modeling claim. Observational equality of the enlarged-state process is insufficient to identify which part of the observed transition came from behavior selection and which part came from world response.

7.3 Baseline dynamical equivalence need not imply constitutive equivalence



Baseline dynamical equivalence does not imply constitutive equivalence.

Figure 4: Counterexample behind Theorem 7.3. Two factorizations induce the same baseline Markov kernel but react differently to the same action-selection intervention.

Theorem 7.3 (Constitutive non-invariance under baseline equivalence). There exist two factorizations with the same baseline joint kernel and the same Exact GR classification such that a matched action-selection intervention is non-constitutive in one factorization and uniformly constitutive in the other.

Proof. Use the two factorizations from Theorem 7.2 and identify the joint state with $Y = \{\star\} \times \{0, 1\}$. Let $B = B_0 = Y$, let $h(\star, x) = x$ on all of Y , let $H = \{0, 1\}$, let $\nu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$, let $B_1 = B_0$, use counting measure as the reference measure, and take d_ψ to be absolute distance on $Z_\psi = [0, 1]$. The baseline kernel is i.i.d. Bernoulli(1/2) on the world coordinate after the initial state, so B is exactly invariant and the empirical occupation measure converges almost surely to ν by the strong law. Define the total property map $\psi : \Delta(\Omega) \rightarrow [0, 1]$ by

$$\psi(\mu) = \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \int x_t d\mu,$$

where x_t is the world coordinate at time t . On the product path space over a finite discrete state space, each finite- T cylinder functional is continuous; therefore ψ , as a limsup of continuous real-valued functionals, is Borel. Baseline $\psi = 1/2$ from either initial state. Apply the matched action-selection intervention J setting $\alpha(\cdot | \star, x) = \delta_1$. In Factorization A, P_A ignores the action, so K and ψ remain unchanged; the action-selection component is not constitutive. In Factorization B, P_B sets $x' = a$, so after the intervention the process reaches $x = 1$ after at most one transition and $\psi = 1$. The d_ψ -gap is therefore 1/2 uniformly over B_1 . Hence the same baseline K supports different constitutive classifications. Q.E.D.

7.4 Intervention-compatible invariance

The previous theorem does not make constitutive classification arbitrary. It identifies the extra information that must be fixed. Two factorizations are intervention-compatible relative to J_1 and J_2 when they have the same baseline K and their matched interventions also induce the same post-intervention kernel.

Proposition 7.4 (Intervention-compatible invariance). If $K_1 = K_2$ and $K_1^{J_1} = K_2^{J_2}$, and the same ψ , B_1 , descriptor specification, and grounded relevance map are used, then a component is constitutive under J_1 in model 1 if and only if the matched component is constitutive under J_2 in model 2. The same holds for uniform constitution with the same d_ψ distance.

Proof. Baseline path laws coincide and post-intervention path laws coincide. The defining inequalities in (15) and (16) are therefore identical. Q.E.D.

This result provides the appropriate representation-invariance standard for PR claims. Baseline K equivalence is sufficient for GR equivalence. PR equivalence additionally requires agreement on the counterfactual intervention semantics; grounded confirmatory equivalence also requires agreement on the declared regime-observation map.

7.4.1 Frozen intervention-family signatures

Let $\mathcal{J}$ be a predeclared intervention family. For a model M write its frozen intervention signature as

$$\text{Sig}_{\mathcal{J}}(M) = (K_M, (K_M^J)_{J \in \mathcal{J}}). \quad (18a)$$

Proposition 7.4a (Intervention-family invariance).

Let M_1 and M_2 use the same state space, frozen regime/property/relevance semantics, and comparison set B_1 . Suppose there is a predeclared target/type-grammar-preserving bijection $\phi : \mathcal{J}_1 \rightarrow \mathcal{J}_2$ satisfying

$$K_1 = K_2, \quad K_1^J = K_2^{\phi(J)} \quad \text{for every } J \in \mathcal{J}_1.$$

Then every matched pointwise constitutive classification, uniform margin, and protocol-scoped negative classification over that family is identical in the two models.

Proof. Equality of each baseline kernel and each matched post-intervention kernel gives equality of the corresponding point-initialized path laws. Applying the same ψ and d_ψ therefore gives identical $\Delta_{\psi, J}(y)$ for every y and J , hence identical infima and classifications. Q.E.D.

The family and the label matching must be frozen ex ante. The bijection preserves the declared intervention target/type grammar (for example, an action-selection intervention is matched to the corresponding action-selection intervention), rather than merely pairing equal induced kernels after inspection. Equality for the baseline and only a selected subset of intervention labels does not justify an equivalence claim for omitted labels; a post-hoc choice of the matching family is a semantic selection step rather than a representation theorem.

7.4.2 Intervention-compatible state quotients

The next result gives sufficient, not necessary, conditions under which state compression preserves the regime layer. It is intentionally stronger than ordinary baseline lumpability because PR status depends on counterfactual kernels and frozen semantics.

Theorem 7.4b (Intervention-compatible quotient preservation). Let M and $\bar{M}$ be well-posed canonical strategic-world models of Section 3 with joint Polish state spaces $Y = S \times X$ and $\bar{Y} = \bar{S} \times \bar{X}$. Let $q_S : S \rightarrow \bar{S}$ and $q_X : X \rightarrow \bar{X}$ be measurable surjections and let the state compression be the type-respecting product map $q = q_S \times q_X : Y \rightarrow \bar{Y}$. Let $\mathcal{J}$ and $\bar{\mathcal{J}}$ be frozen typed intervention families with a predeclared target/type-grammar-preserving bijection $\phi : \mathcal{J} \rightarrow \bar{\mathcal{J}}$; treat the baseline as label 0 and set $\phi(0) = 0$. Write K^J and $\bar{K}^{\phi(J)}$ for the induced baseline or post-intervention kernels of the two canonical models. Suppose:

- (i) for every $J \in \{0\} \cup \mathcal{J}$,

$$K^J(q^{-1}(C) \mid y) = \bar{K}^{\phi(J)}(C \mid q(y))$$

for every y and Borel $C \subseteq \bar{Y}$;

- (ii) $B = q^{-1}(\bar{B})$, $B_0 = q^{-1}(\bar{B}_0)$, and both the original and quotient specifications independently satisfy Assumption 4.1;
- (iii) $h = \bar{h} \circ q$ and the same descriptor target law ν and convergence mode are used;
- (iv) $B_1 = q^{-1}(\bar{B}_1)$, and B_1 and $\bar{B}_1$ independently satisfy the comparison-set admissibility/non-triviality requirement of Section 5.4;
- (v) ψ and $\bar{\psi}$ take values in the same metric property space (Z_ψ, d_ψ) , and on all baseline and frozen-intervention path laws used in the comparison,

$$\psi(P) = \bar{\psi}((q^\infty)_\# P);$$

for grounded claims, $g = \bar{g} \circ q$; and

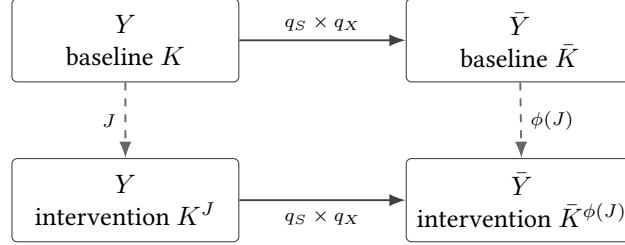
- (vi) every probability law on $\bar{B}_0$ considered by the quotient specification has an admissible lift λ on B_0 with $q_\# \lambda = \bar{\lambda}$.

Then Exact-GR status is preserved between M and $\bar{M}$, and for every $J \in \mathcal{J}$ the corresponding pointwise and uniform PR classifications and margins are preserved under the matched intervention $\phi(J)$. The theorem assumes, rather than reconstructs, the canonical strategic/world factorization and typed intervention/component grammar of the quotient model; the product form of q prevents the quotient itself from relabelling strategic coordinates as world coordinates or conversely.

Proof. The kernel-intertwining identity in (i) implies by induction on cylinder sets that $(q^\infty)_\# P_y^{K^J} = P_{q(y)}^{\bar{K}^{\phi(J)}}$ for every point initial state y and every J , including the baseline. Because $B = q^{-1}(\bar{B})$, exact invariance is equivalent statewise under q . Because $h = \bar{h} \circ q$, the descriptor sequences agree pathwise after projection, so their empirical occupation measures are identical under pushforward. Assumption (vi) lifts every admissible quotient initial law to an admissible original initial law, giving the required occupation-law convergence for the full quotient basin; the converse direction follows by

pushing original initial laws forward. Thus the GR gate is preserved, subject to the separately imposed non-triviality gates in (ii). For PR, assumption (v) makes each baseline and matched post-intervention property value equal to its quotient counterpart in the same metric property space. Since $B_1 = q^{-1}(\bar{B}_1)$, q is onto $\bar{B}_1$, and both comparison sets satisfy Section 5.4, $\Delta_{\psi,J}(y) = \bar{\Delta}_{\bar{\psi},\phi(J)}(q(y))$ and the infimum over B_1 equals the infimum over $\bar{B}_1$. Pointwise constitution and uniform margins therefore coincide. Grounded status follows from $g = \bar{g} \circ q$. Q.E.D.

The assumptions are material. Untyped or baseline-only lumpability does not preserve PR status if the compression mixes strategic/world coordinates or if some matched post-intervention kernel fails to descend; a quotient can also destroy the regime claim if B , B_0 , h , ψ , the comparison-set guard, or the relevance semantics fail to factor or remain admissible. The non-triviality conditions are imposed separately because topological interior, reference-measure positivity, descriptor diversity, and the Section 5.4 comparison-set condition need not be invariant under an arbitrary quotient map.



The square must commute for baseline and every frozen matched J , and the canonical typed semantics plus B , B_0 , h , ψ , g must remain compatible.

Figure 5: Intervention-compatible typed quotient requirement. The state compression is type-respecting, and baseline lumpability alone is insufficient for PR preservation.

7.5 Nuisance-state padding and grounded constitution

The stress-test motivation is an augmentation $\tilde{Y} = Y \times R$ in which an auxiliary bookkeeping coordinate $r \in R$ is updated by the strategic generator but does not alter the substantive regime observation. If a constitutive property is allowed to inspect the whole augmented path, an intervention on the auxiliary coordinate can manufacture $\text{PR}_{\text{general}}$ status even when the world process is unchanged. The grounded subclass prevents that result unless the auxiliary coordinate was explicitly included in the frozen relevance map.

Proposition 7.5 (Nuisance-padding exclusion under a fixed relevance map). Let $\tilde{Y} = Y \times R$ and let $\tilde{g}(y,r)=g(y)$. Suppose a baseline model and its intervention have the same projected g -path laws before and after adding or modifying the auxiliary coordinate r . Then every $\tilde{g}$ -grounded property ψ has the same baseline and post-intervention values as in the unpadded representation. In particular, changing r alone cannot create grounded constitutive status. If an application intends r to be substantively regime-defining, r must be included in $\tilde{g}$ before the counterfactual is evaluated.

Proof. By definition, a $\tilde{g}$ -grounded property factors through the pushforward path law $(\tilde{g}^\infty)_\# \mu$. The auxiliary coordinate is removed by $\tilde{g}$, and the assumed projected baseline and post-intervention laws are unchanged. Therefore the property values and all d_ψ gaps are unchanged. Q.E.D.

8. Canonical Examples

8.1 A periodic Generated Regime without an equilibrium primitive

Let $S = X = A = \{0, 1\}$ and take every action to be feasible at every state. Define $\alpha(a = s \mid s, x) = 1$, $P(x' = 1 - a \mid s, x, a) = 1$, and $U(s' = 1 - s \mid s, x, a, x') = 1$. Consider the two-state subset

$$B = B_0 = \{(0, 0), (1, 1)\}. \quad (19)$$

From (0,0), the process chooses $a=0$, sets $x' = 1$, and updates $s' = 1$, reaching (1,1). From (1,1), it reaches (0,0). Thus B is exactly invariant and the joint process has period two. Let $H=Y$ with the discrete topology and let $h : Y \rightarrow H$ be the identity map. Then

$$\nu = \frac{1}{2}\delta_{(0,0)} + \frac{1}{2}\delta_{(1,1)}. \quad (20)$$

is the almost-sure limiting occupation law from either initial state. With the counting reference measure, Assumption 4.1 is satisfied. This is an Exact GR even though no equilibrium certificate is a primitive of the construction. The example demonstrates that equilibrium certification is not logically necessary to the regime-classification layer; it does not claim

that the periodic process could not also arise from some separately specified equilibrium model, nor that periodic dynamics are new.

Proposition 8.1. The tuple defined by (19)-(20) is an Exact GR under almost-sure weak occupation convergence.

Proof. The transition is deterministic and alternates between the two points of B, so exact invariance is immediate. The empirical frequency of each point differs from 1/2 by at most 1/T, giving weak convergence of the empirical occupation measure to ν from both initial states. Q.E.D.

8.2 A discrete strategy-world feedback loop

A slightly different interpretation makes the feedback semantics explicit. Let X_t indicate a depleted (0) or replete (1) resource state. Let the strategic generator choose a low-impact action when the resource is depleted and a high-impact action when it is replete. Let the low-impact action restore the resource and the high-impact action deplete it. The resulting deterministic loop alternates between behavior and resource conditions. This is a discrete toy analogue of the bidirectional strategy-environment coupling formalized in feedback-evolving games; it is not intended as a discretization theorem for any particular continuous-time model.

Continuous-time eco-evolutionary systems can be compared to the canonical framework by considering a time- Δ skeleton when the flow or Markov process is well defined. However, exact equivalence between continuous-time persistence and skeleton persistence requires additional conditions: a trajectory may leave and re-enter B between sample times. Continuous-time GRs therefore belong in a separate extension rather than being smuggled into the discrete core.

8.3 Same baseline regime, different constitution

Theorem 7.3 is also a substantive example. Both models generate the same i.i.d. Bernoulli world process, the same invariant region, and the same limiting occupation law. A baseline-only regime analysis cannot distinguish them. Once the same action-selection intervention is applied, one system is unaffected while the other becomes absorbed. The difference is not an additional attractor concept; it is a statement about which typed mechanism is constitutive of the baseline regime.

9. Regime Profiles and Feedback Diagnostics

9.1 Separate generator class from regime class

A central design rule is that generator type and regime type are different axes. An equilibrium generator can produce a changing world state. A learning generator can converge to a fixed point. A replicator generator can generate a cycle. A stochastic policy can induce an invariant distribution. The paper therefore avoids using “equilibrium regime” and “non-equilibrium regime” as exhaustive dynamical categories.

Axis	Illustrative labels
Generator class	equilibrium-certified; best-response; fictitious-play; reinforcement; no-regret; evolutionary; fixed policy; exogenous
Regime locus	world; strategic; coupled
Asymptotic form	fixed; periodic; recurrent; limiting-occupation; invariant-factor (when verified); balanced-growth; contracting; target-approaching
Persistence status	exact; $(L, \eta) - quasi$; QSD-certified quasi-regime; metastable extension
Constitution	generator-constitutive; structurally constitutive; jointly constitutive; not established
Feedback	reinforcing; limiting; mixed/self-regulating; not identified

Table 3. Multi-axis Generated Regime profile. Labels require explicit mathematical or intervention-relative definitions in applications.

9.2 Typed feedback graph

For interpretation, a model may carry a signed directed graph whose nodes are components of S and X. Edges $S \rightarrow X$ represent model-implied effects of strategic variables or actions on world dynamics; edges $X \rightarrow S$ represent effects of world state on future action selection or strategic updating. A reinforcing cycle amplifies a predeclared regime property, a limiting cycle attenuates it, and mixed loops can produce bounded persistence or oscillation. These graphs are model-relative bookkeeping unless separate identification assumptions justify a causal interpretation.

The typed graph is useful because an enlarged-state dynamical system treats all coordinates symmetrically. The graph instead records the analyst's structural claim about where behavior is generated and where consequences enter. Theorem 7.2 warns that this claim cannot generally be inferred from K alone.

10. Analytical Implications

10.1 Equilibrium becomes a generator certificate, not a regime primitive

Under the generalized framework, equilibrium retains an important role but a narrower one. It certifies a particular strategic generator or action rule. The regime layer then asks what process that certified behavior produces when composed with the world transition. This recovers the original EGR logic exactly while making clear that learning, evolutionary, or exogenous generators can feed the same regime layer after their own behavioral assumptions are specified.

10.2 The object is analyst-relative by design

A GR is not the unique "true regime" mechanically extracted from a process. It is a declared classification object. This is a feature rather than a defect, provided the declaration is *ex ante* and nontrivial. Standard dynamical systems may identify maximal invariant sets, chain-recurrent components, attractors, or ergodic measures intrinsic to a chosen process. GR analysis instead asks whether a specific region, descriptor, and occupation property relevant to a substantive question is satisfied. The cost is analyst dependence; the discipline is predeclaration, held-out validation, and explicit convergence conditions.

10.3 Why the typed factorization earns its keep

If the objective were only to classify the baseline trajectory, the factorization could be discarded after forming K . Theorem 7.3 shows the additional value: baseline dynamical equivalence does not imply constitutive equivalence. A modeler may care not only that a regime persists but whether it persists because agents keep selecting a behavior, because the world mechanically amplifies that behavior, because the learning rule reconstructs it after shocks, or because several mechanisms are jointly necessary. These are typed counterfactual questions.

10.4 Implications for empirical work

The non-identification theorem imposes a hard claim boundary. Observed state trajectories cannot by themselves generally identify α , U , and P , and even full observational action support does not guarantee that an observational conditional distribution is the causal world mechanism preserved under an intervention. Empirical PR claims therefore require intervention-relevant identification and an explicit account of what is observed, estimated, structurally assumed, or externally certified. The required evidence may come from structural restrictions, experiments, natural experiments, valid instruments, sufficient-state adjustment under justified assumptions, direct measurements, engineering or institutional mechanism knowledge, or another defensible causal design. Without that information, one may establish a GR relative to an estimated K or path law while remaining agnostic about strategic constitution.

This distinction mirrors the earlier EGR warning that a signed feedback graph is not automatic causal identification and that interventions must state what changes and what remains fixed. The generalized framework makes the same discipline even more important because generator and world mechanisms are separately typed. Confirmatory empirical use must additionally audit intervention-reachable support, causal modularity, and provenance; model-imputed off-support quantities cannot serve as the sole evidence validating the same model that imputed them. Ambiguity is a valid result and should be preserved rather than resolved by choosing a convenient structural completion.

10.5 Three-layer confirmatory architecture

The confirmatory architecture separates the formal mathematical object from the semantic counterfactual specification and the empirical certificate. A failure in the empirical layer downgrades the application certificate; it does not rewrite the mathematical GR/PR object.

Layer	Contents	Failure consequence
A - Mathematical regime core	Canonical process, GR specification, Exact GR gate, generalized PR definition, representation/intervention theorems.	Formal GR/PR status is determined only by the mathematical assumptions and declared intervention.
B - Frozen semantic counterfactual specification	State decomposition and sufficiency; B , B_0 , h , ν , c ; relevance map g ; property ψ ; B_1 ; component grammar; J ; hard target closure; semantic transport rules.	A semantic ambiguity or post-hoc switch blocks confirmatory use unless the search is validly adjusted or held out.

Layer	Contents	Failure consequence
C - Empirical certificate	Rooted provenance; intervention-reachable support; causal identification; identified model set; outer joint structural/statistical uncertainty; selection/multiplicity/sequential procedure; application certificate tuple.	Failure downgrades or invalidates the empirical certificate while leaving Layer A unchanged.

Table 4. Three-layer architecture for confirmatory use. The certification layer is an annotation on an application, not a new mathematical regime class.

For confirmatory use, Layer B is frozen as one joint pipeline identifier rather than as a collection of separately mutable choices. The identifier should bind g , the component grammar, J , $\psi/outcome$ definition, hard structural conventions, model-set construction, and inferential procedure. If the analyst searches across the Cartesian product of these choices, confirmation requires a valid held-out or selection-adjusted analysis that accounts for the search.

The provenance record in Layer C should be an acyclic rooted derivation graph rather than a flat list. Every empirical assumption must ultimately trace to admissible external evidence, a formally declared identity/axiom, or an independently certified upstream result. Circular support does not become valid merely because every node has a citation to another node in the same cycle.

10.6 Partial identification and outer uncertainty sets

When evidence identifies a provenance-valid set of structural models M rather than one model, confirmatory PR classification must quantify over the full set rather than a representative, average, majority, posterior-majority, or convenient witness. For a declared protocol, robust pointwise PR requires a nonzero constitutive effect for every $m \in M$ and every $y \in B_1$. Robust uniform PR additionally requires one common positive d_ψ gap over the same joint set.

Applications should separate three questions: (i) PR status - whether constitution is nonzero throughout the admissible model/state set; (ii) direction - whether the sign is common; and (iii) uniform gap - whether one $\delta_\psi > 0$ works over the entire admissible set. Robust PR can therefore coexist with direction ambiguity, and pointwise robustness need not imply a common positive uniform gap.

If the identified set is estimated from finite data, structural and sampling uncertainty must be combined into one valid outer joint model/effect set before classification. Intersecting per-model confidence sets, averaging effects, or classifying uncertainty layers separately can manufacture certainty. When a valid covered outer set still contains a zero-constitution case, the statistical status is STATISTICALLY_UNRESOLVED. Failure to statistically certify PR is not, by itself, evidence of nonconstitution.

The statistical procedure must have justified coverage for the actual inferential object. Where relevant this includes simultaneous coverage over B_1 , multiple interventions, model-set endpoints, data-dependent selection, and sequential monitoring. Fixed-sample marginal intervals are not automatically valid after optional stopping or post-hoc intervention search; multiplicity correction cannot repair an intrinsically under-covering base procedure.

10.7 Orthogonal application certificate

The application-status vocabulary is not a mutually exclusive enumeration. A confirmatory report should retain an orthogonal certificate tuple so one dimension cannot erase another. A compact headline may be derived for presentation, but the underlying tuple should remain auditable.

Table 5: Orthogonal application-certificate tuple. These are reporting annotations, not additional GR/PR theorem classes.

Dimension	Illustrative values
Validity	VALID; INVALID
Epistemic mode	FORMAL_THEORETICAL; EMPIRICAL_CONFIRMATORY; EMPIRICAL_EXPLORATORY; ASSUMPTION_CONDITIONAL
Selection status	FROZEN_REGISTERED; HELD_OUT_OR_SELECTION_ADJUSTED; POST_HOC_EXPLORATORY
Identification status	POINT_IDENTIFIED; SET_IDENTIFIED; STRUCTURALLY_UNRESOLVED
Statistical status	NOT_APPLICABLE; RESOLVED_AT_DECLARED_COVERAGE; STATISTICALLY_UNRESOLVED
Provenance status	SUPPORTED; INSUFFICIENT; NOT_APPLICABLE
Substantive PR status	ROBUST_PR; ROBUST_NONCONSTITUTIVE_UNDER_FROZEN_PROTOCOL; PR_STATUS_AMBIGUOUS; NOT_EVALUATED
Direction status	POSITIVE; NEGATIVE; DIRECTION_AMBIGUOUS; NOT_APPLICABLE
Uniform-gap status	ESTABLISHED; UNRESOLVED; NOT_APPLICABLE

11. Extensions Beyond the Canonical Discrete-Time Core

11.1 Continuous time

A continuous-time version replaces the discrete joint kernel by a flow, semigroup, stochastic differential equation, or controlled generator on $S \times X$, while retaining a typed decomposition of strategic and world dynamics. For deterministic models one natural form is

$$\dot{S}_t = F_S(S_t, X_t), \quad \dot{X}_t = F_X(S_t, X_t). \quad (21)$$

with F_S interpreted as strategic-generation/update dynamics and F_X as world dynamics. Eco-evolutionary games fit this template directly. The corresponding GR definition would use forward invariance, occupation measures, recurrent sets, or other continuous-time asymptotic descriptors. Because continuous trajectories can exit a region between discrete sample times, continuous-time exact persistence should be defined directly rather than inferred from a time-skeleton.

11.2 Differential inclusions and set-valued generators

Best-response correspondences, discontinuous adaptation, and some bounded-rational processes are more naturally represented by differential inclusions or set-valued dynamics. Benaïm, Hofbauer, and Sorin (2005) already supply an appropriate asymptotic substrate through internally chain-transitive sets and attractors. A continuous-time PR extension should reuse that mathematics and add only the typed regime and intervention semantics.

11.3 Nonautonomous and random generators

If the generator or world law changes exogenously with time, the appropriate baseline object may be a nonautonomous process or cocycle rather than a stationary Markov kernel. Kloeden and Rasmussen (2011) develop process and skew-product formulations, pullback attractors, switching systems, and random dynamical systems. A future nonautonomous PR should therefore index the strategic generator and world law by time or an external driving process, for example $\mathcal{G}_t$ and P_t , and replace stationary occupation assumptions where necessary. This extension lies outside the discrete-time framework developed here.

11.4 Partial observability and belief states

Partially observed strategic systems can often be made Markov by augmenting S with sufficient beliefs, filters, or internal information states. When no finite- or countable-dimensional sufficient statistic exists, the strategic state may itself be a probability measure or other infinite-dimensional object. The Polish-space core accommodates many such cases, but application-specific existence and measurability conditions must be checked.

12. Falsifiability, Failure Modes, and Claim Boundary

12.1 Ways the framework can fail

The framework is intentionally falsifiable at several levels.

- Well-posedness failure: α , U , or P is not measurable or does not preserve feasibility, so the canonical process is undefined.
- Regime-specification failure: B , B_0 , h , ν , or the convergence mode is chosen after observing the evaluated trajectory or is trivial under Assumption 4.1.
- Persistence failure: the declared B is not exactly invariant, so the object is not an Exact GR.
- Occupation failure: the empirical descriptor measures do not converge to the declared ν under the stated mode for all admissible initial laws.
- Protocol-relative constitution failure: under the frozen intervention protocol, the proposed strategic component can be intervened upon without changing ψ on the declared comparison set. This excludes constitution under that protocol; it does not by itself exclude every other admissible intervention family.
- Identification failure: available evidence establishes only K or a path law, or leaves intervention-relevant mechanism rows/model completions unresolved, so the decomposition/counterfactual required for a strategic constitutive claim is not identified.
- Numerical pseudo-exactness: floating-point equality or a tolerance is used to promote a leaky process to Exact GR; exact claims require analytic, symbolic/rational, certified-interval, proof-object, or equivalent exactness support.
- Statistical-certification failure: the confidence region undercovers the actual joint inferential object, ignores multiplicity or optional stopping, or a zero-containing valid set is treated as resolved PR status.

- Partial-identification overreach: admissible models disagree about constitution but the application reports a representative, average, majority, or single-witness classification as if status were identified.
- Provenance failure: an empirical assumption is supported only by a circular/rootless derivation graph, post-hoc hard/soft labeling, or a model-imputed quantity that is used to validate the same model.
- Intervention-support/modularity failure: the intervention reaches unsupported rows or relies on held-fixed causal mechanisms whose stability under J lacks adequate identification/provenance.
- Empirical-admissibility failure: a formally typed intervention violates a predeclared hard structural relation outside its declared target, or the target is not closed under the hard dependency relation.
- Relevance/semantic-selection failure: g , the component grammar, ψ , J , or another semantic counterfactual element is chosen after seeing the result without a valid held-out or selection-adjusted procedure.
- State-sufficiency failure: the observed or aggregated state is treated as Markov without a valid lumpability/sufficiency argument; hidden states within one aggregate label can therefore induce different continuation laws.
- Reduction failure of novelty: if a proposed application uses only baseline attractors or invariant sets and no typed intervention semantics, standard dynamical-systems language may be sufficient and PR terminology may add nothing.
- QSD overclaim: a QSD is invoked without proving existence for the killed kernel, or uniqueness is assumed where several QSDs can coexist.
- Perturbation overreach: a finite-horizon perturbation bound is extrapolated to stationary or infinite-path behavior without the ergodic, contraction, spectral, or killed-process assumptions needed for such a step.
- Margin-boundary failure: a robustness claim uses $\kappa \geq \varepsilon_0 + \varepsilon_J$ rather than the strict certificate $\kappa > \varepsilon_0 + \varepsilon_J$, or treats pointwise nonzero effects as a positive uniform margin.
- Intervention-family omission: an equivalence claim matches only a subset of frozen intervention kernels, or chooses the family after observing which counterfactuals agree.
- Quotient overreach: a state quotient is only baseline-lumpable while an intervention kernel, regime set, descriptor, property map, relevance map, or admissible initial-law class fails to descend.
- Rigidity-domain failure: a candidate scalar-defect, finite-detection, first-bad, singular-compatibility, or witness theorem is applied outside the subclass that supplies its hypotheses.

12.2 Nonclaims

The canonical formulation does not establish equilibrium existence for arbitrary games; universal convergence of learning; empirical causal identification from observational trajectories; general recurrence or ergodicity of K ; existence of limiting occupation laws for arbitrary generators; equivalence between metastability and exact persistence; or a complete continuous-time, nonautonomous, or differential-inclusion PR theory. These must be supplied by the underlying model or the neighboring mathematical literature. It also does not imply that a relevance map, component decomposition, causal factorization, identified model set, or inferential procedure is uniquely recoverable from baseline trajectories; nor does an application certificate supplied by Section 10 convert assumption-dependent evidence into point identification.

12.3 Novelty standard

The relevant novelty test is functional rather than terminological. Mature adjacent literatures already contain most mathematical ingredients used here. Labelled Markov-process theory supplies bisimulation and behavioral metrics for label-indexed kernel families; causal-abstraction theory studies preservation of interventional predictions across representations; killed-process theory supplies QSD and exit-time machinery; and Markov perturbation theory supplies kernel-to-law sensitivity bounds under explicit regularity assumptions. Accordingly, this paper does not claim novelty for those ingredients, nor for the broad observation that baseline equivalence can coexist with intervention nonequivalence. The framework-specific claim is narrower: a typed strategic/world regime layer that combines *ex ante* GR semantics, predeclared constitutive generator interventions, grounded confirmatory semantics, exact persistence calculus, quantitative margin certification, and sufficient family-compatible quotient conditions in one portable object. That assessment remains provisional and should be revised if a prior framework is shown to provide the same combined function.

13. Conclusion

The original EGR question was asked after equilibrium: once a strategic policy is certified, what persistent world does it generate? The generalized formulation shows that equilibrium is not a primitive of the regime layer. It is one possible certificate for a strategic generator. A learning rule, evolutionary protocol, fixed policy, or other sufficiently specified behavioral process can generate the same kind of regime-classification problem.

At the same time, the framework does not replace dynamical-systems or Markov-process mathematics. The joint process is an ordinary Markov process on an enlarged state, and its invariant sets, recurrence, occupation measures, killed-process

behavior, or metastability should be analyzed with established tools. The additional object is typed and intervention-relative. A Generated Regime classifies a declared persistent process. A Permansson Regime adds the claim that a specific component of the strategic generator is constitutive of a defining regime property under an admissible intervention.

The quantitative layer makes this separation sharper. Polish descriptor spaces broaden the ambient class; killed kernels provide exact finite-persistence and exit-time identities; QSDs furnish an optional finite-horizon certificate; constitutive margins quantify robustness to bounded property-output error; frozen intervention-family signatures extend representation equivalence; and the quotient theorem gives sufficient conditions under which GR/PR semantics survive type-respecting state compression. These results do not replace the application disciplines of relevance, component grammar, causal modularity, support, provenance, identification, or statistical uncertainty.

The central result is therefore the separation between baseline dynamical equivalence and constitutive equivalence. Two models may generate the same baseline joint process and the same long-run regime while responding differently to strategic intervention. Conversely, when baseline and matched intervention kernels and the relevant semantics agree, the corresponding constitutive classifications agree. The Permansson layer preserves the typed intervention information needed to ask not only what regime a process generates, but which strategic mechanisms are constitutive of that regime.

14. Appendix A - Compact Theorem and Definition Ledger

Table 6: Core GR/PR theorem and definition ledger.

Item	Core statement
Joint-process construction	α , P , and U compose into $K_{\mathfrak{G}, P}$; Ionescu–Tulcea gives a unique path law.
Exact GR	Ex ante specification + exact invariance of B + limiting descriptor occupation law for all λ supported on B_0 .
Generalized PR	Exact GR + at least one strategic-generator component constitutive under a fixed admissible $J^{\mathfrak{G}}$.
EGR embedding	Use $S_E = \mathbb{N}_0 \times (A \cup \{\perp\})$ to record clock and previous action; $\alpha = \delta_{\pi^*}$, U records the realized action, and the X -marginal equals the Paper-I equilibrium-induced kernel.
Conservative PR extension	Paper-I date/state policy interventions map to clock-indexed action-selection interventions; under a well-defined frozen Paper-I descriptor/property convention, baseline and counterfactual executions and constitutive comparisons are preserved.
Baseline representation invariance	Same K + same Σ + same reference-measure convention $\Rightarrow$ same GR classification.
Factorization non-identification	Same K does not imply same $\mathfrak{G}$ or P .
Constitutive non-invariance	Same baseline K can yield different PR status under the same typed intervention.
Intervention-compatible invariance	If both baseline and matched post-intervention K coincide, constitution classification coincides.

Table 7: Quantitative and representation-theoretic results.

Item	Core statement
Polish descriptor-space generality	H may be any Polish descriptor space equipped with a bounded compatible metric; d_{BL} still metrizes weak convergence. No existence/tightness conclusion is automatic.
Exact killed-kernel persistence	For $y \in B$, $P_y(\tau_B > n) = K_B^n \mathbf{1}_B(y)$, and $E_y[\tau_B] = \sum_{n \geq 0} K_B^n \mathbf{1}_B(y)$ as an extended identity.
QSD-certified quasi-regime	An (L, η) -quasi-regime may carry an additional QSD certificate $\mu_B K_B = \theta_B \mu_B$; then survival from μ_B is θ_B^n and the conditional law given survival is μ_B . The QSD equation does not replace the uniform finite-persistence gate, and existence/uniqueness require separate hypotheses.
Constitutive robustness margin	Uniform property-output errors $\varepsilon_0, \varepsilon_J$ imply $ \tilde{\kappa} - \kappa \leq \varepsilon_0 + \varepsilon_J$; $\kappa > \varepsilon_0 + \varepsilon_J$ certifies persistence of uniform constitution.
Intervention-family invariance	A frozen target/type-grammar-preserving matching of the indexed intervention family, with equal baseline and matched K^J , implies identical corresponding effect profiles, uniform margins, and protocol-scoped classifications.
Intervention-compatible quotient preservation	Between two canonical typed models under a type-respecting product map q , if the baseline and every frozen matched intervention kernel intertwine, both comparison sets remain admissible, and B, B_0, h, ψ, g plus initial-law lifting are compatible, GR/PR classification descends and lifts.

The relevance, intervention, provenance, identified-set, and statistical rules are application/certification disciplines surrounding the mathematical core. The persistence, robustness, family-equivalence, and quotient results above do not alter the set-theoretic hierarchy of GR, PR_{general} , PR_g , or the transported Paper-I subclass. The specialized finite-certificate and rigidity material in Appendix E lies outside this core ledger.

15. Appendix B - Proof Notes and Technical Clarifications

15.1 Measurability of the composed kernel

For bounded measurable $f : Y \rightarrow \mathbb{R}$, define the Markov operator

$$(Kf)(s, x) = \int_A \int_X \int_S f(s', x') U(ds' \mid s, x, a, x') P(dx' \mid s, x, a) \alpha(da \mid s, x). \quad (22)$$

Successive kernel integration preserves measurability. Taking $f = \mathbf{1}_E$ yields (6). This is the standard kernel-composition argument used in controlled Markov processes and does not require a new existence theorem.

15.2 Why the descriptor is global

The earlier EGR definition used $h : B \rightarrow H$ because exact equilibrium persistence keeps the baseline path inside B . In the generalized intervention setting, the counterfactual process may leave B precisely because the intervention destroys a defining regime property. Defining h globally avoids an unnecessary extension step. For the EGR embedding, any constant Borel extension outside B preserves all baseline EGR claims. For the conservative Permansson theorem, however, if a Paper-I property functional evaluates h after an intervention exits B , the extension must be part of the frozen intervention/descriptor convention; the theorem does not manufacture an ex post value for an otherwise undefined Paper-I descriptor.

15.3 Why B_1 has at least two initial states

A constitutive claim evaluated at a single hand-picked initial condition can be fragile and can collapse into a trajectory-specific sensitivity statement. The Paper-I wording therefore requires at least two initial states. Because full point-initialized path laws include their deterministic time-zero coordinate, “distinct path laws” is no stronger than “distinct states” unless a continuation-law qualifier is added. Distinct shifted continuation laws remain a useful diagnostic, but the framework does not treat them as a universally strongest guardrail. For grounded confirmatory work, an optional stronger audit can compare g -projected baseline/intervention response profiles: two states are substantively distinguishable when their grounded path laws differ at baseline or under the frozen counterfactual. Even this is a diagnostic rather than a universal gate, because latent types, beliefs, and learning states can pool under baseline observations yet respond differently under J . None of these diversity diagnostics substitutes for a defensible predeclared relevance map. Uniform constitution strengthens the substantive comparison by imposing a positive d_ψ separation over B_1 . Basin-only or pooled-baseline classifications should be described transparently rather than rejected by a rule that would also exclude legitimate contraction and hidden-state models.

15.4 Why exact GR requires invariance on all of B

If exact persistence were required only from B_0 , the analyst could place unreachable leakage states inside B and still label B a regime region. Requiring $K(B | y) = 1$ for every $y \in B$ makes B genuinely invariant. The initial basin B_0 then controls which subset of the invariant region is claimed to share the declared occupation law.

15.5 Why equality of baseline K is not enough for PR equivalence

A factorization defines a counterfactual grammar. Theorem 7.2 shows that many grammars can induce the same baseline K . Permansson constitution is therefore not a function of K alone; it is a function of the baseline process plus the typed intervention map. Proposition 7.4 states the appropriate invariance condition for matched interventions. Proposition 7.5 adds the representation guard: a confirmatory property is evaluated only through a frozen relevance map g , so nuisance coordinates omitted from g cannot manufacture grounded PR status. This does not make substantive relevance mathematically automatic; it forces the analyst to declare which coordinates are allowed to matter before the counterfactual is seen.

15.6 Why empirical admissibility is separate from typed intervention

A typed intervention is a mathematical surgery on declared kernels. That is sufficient to define the formal counterfactual but not to establish that the same surgery is empirically meaningful in a real system. Hard structural identities can force the effective target to expand; intervention-reachable rows may lie outside baseline support; and an observational conditional distribution may differ from the causal mechanism that remains fixed under intervention. These are evidence and model-identification questions, so they are treated as certificate-layer requirements rather than alterations to Definition 5.1.

15.7 Why negative claims are protocol-scoped

Permansson constitution is defined relative to a property map, intervention protocol, and comparison set. A null result for one J therefore establishes only nonconstitution under that frozen protocol. A global absence claim requires exhaustion of a predeclared admissible intervention family or an independent theorem excluding the family. This scope rule prevents failure of one intervention from being interpreted as the much stronger statement that no strategically constitutive intervention exists.

15.8 Why uncertainty is classified through one outer set

Structural ambiguity and sampling uncertainty can interact. The safe object for confirmatory classification is therefore one outer joint set containing every model/effect configuration retained by the provenance-valid structural restrictions and the declared statistical coverage procedure. Intersecting separately valid per-model confidence sets, averaging model

effects, or classifying structural and sampling uncertainty in separate passes can remove zero-effect possibilities without justification. Robust PR, direction, and uniform-gap claims are assigned only after this outer set has been constructed.

15.9 Polish descriptors and bounded-Lipschitz convergence

Compactness was stronger than the core definition required. A Polish space admits a complete separable compatible metric ρ ; truncating it to $\min\{1, \rho\}$ preserves the topology and completeness while making it bounded. The associated bounded-Lipschitz/Fortet–Mourier metric metrizes weak convergence of probability measures. This justifies the replacement in Section 4.1. It does not supply uniform tightness for an arbitrary family of measures, so the existence of ν remains part of Exact-GR certification rather than a topological consequence (Dudley, 2002).

15.10 Exact killed-kernel indexing

The convention $\tau_B = \inf\{t \geq 0 : Y_t \notin B\}$ means that, from $y \in B$, survival beyond n is exactly retention at times 1 through n . This is why K_B^n rather than K_B^{n+1} appears in (11b). The $n = 0$ identity is one, the induction step integrates only over B , and the Green-series formula follows from the tail-sum identity. This convention is fixed throughout the paper to prevent off-by-one ambiguity.

15.11 QSD claims are conditional

Equation (13a) is an eigenmeasure relation for the sub-Markov kernel. It implies geometric survival and conditional stationarity algebraically, but it neither proves that such a probability measure exists nor that it is unique. Those are separate killed-process theorems. The paper therefore uses QSD only as an additional certificate for an already established finite-horizon quasi-regime and never treats QSD, Yaglom limit, and quasi-ergodic law as synonyms. The eigenmeasure equation alone does not establish the uniform finite-persistence gate in (11).

15.12 Why the perturbation theorem is only a bridge

Proposition 5.2 is pure metric geometry: it begins after errors in property-output space have already been bounded. Obtaining those errors from a kernel perturbation requires additional assumptions. Proposition 5.4 supplies a finite-horizon bound with no ergodicity requirement; long-run bounds must instead be imported from the appropriate uniform-ergodic, contraction, spectral, or killed-process theory. This modularity prevents a local one-step perturbation statement from being overextended to infinite-horizon behavior.

15.13 Why a full intervention signature is required

A family claim is a conjunction over the frozen intervention labels. Equality of the baseline and one post-intervention kernel proves equivalence only for that matched label. If an omitted label produces different K^J , its effect profile can differ even though all checked labels agree. Proposition 7.4a therefore requires the entire predeclared indexed signature relevant to the claim and a target/type-grammar-preserving label matching; selecting either the family or its matching after inspection is outside the representation theorem.

15.14 Why quotient preservation needs more than lumpability

Ordinary lumpability concerns one kernel. A PR claim concerns a baseline path law, one or more intervention path laws, and frozen semantics. Theorem 7.4b therefore starts with two well-posed canonical typed models and a type-respecting product state map, requires a target/type-grammar-preserving intervention matching, kernel intertwining for every frozen label, a common property metric, and factorization of B, B_0, h, ψ and, when applicable, g . The comparison sets must independently satisfy the Section 5.4 admissibility guard. The initial-law lifting assumption is included because Exact GR quantifies over every admissible initial law in the quotient basin; pointwise kernel intertwining alone does not automatically provide a measurable lift for every quotient law. Non-triviality is checked separately on each specification because interior, reference-measure positivity, descriptor diversity, and comparison-set admissibility are not invariant under arbitrary quotients.

16. Appendix C - Suggested Empirical and Computational Workflow

For applied work, the following sequence keeps the claim boundary auditable:

1. Declare the typed state decomposition $S \times X$. Establish Markov sufficiency/lumpability for any observed or aggregated state, augment the state when necessary, or fail closed on Exact-GR certification.
2. Specify α, P , and U or an estimable/identified approximation, and distinguish observed quantities, estimated quantities, formal identities, and structural assumptions.
3. Freeze B, B_0, h, ν, c , and any reference-measure convention before confirmatory evaluation.
4. Verify exact invariance with an exact mathematical argument. For finite-horizon analysis, construct K_B and compute or bound $K_B^L \mathbf{1}_B$ rather than relying only on a one-step q^L diagnostic. If a QSD certificate is claimed, establish the

eigenmeasure relation and any existence, uniqueness, Yaglom-limit, or quasi-ergodic claims separately; a QSD equation does not replace the finite-persistence gate. Do not promote tolerance-based or raw floating-point equality to Exact GR; otherwise report finite-horizon/quasi/metastable evidence.

5. Establish descriptor occupation convergence under the declared mode. Verify basin-reachable descriptor nondegeneracy for confirmatory certification and report separately whether forward descriptor richness is satisfied.
6. Define and audit the relevance map g . Verify factorization of B and h and grounding of ψ ; search strict coarsenings when feasible; report non-unique sufficient maps; and rerun defensible alternatives when semantics are not unique.
7. Freeze one joint semantic pipeline identifier containing g , ψ , B_1 , the component grammar, J , hard structural target/transport conventions, model-set construction, and the inferential procedure. If selection occurs, use held-out or valid selection-adjusted confirmation.
8. Conduct a component/intervention audit. Record whether the target is atomic, compound, or joint; test substantively meaningful granularity changes; transport interventions semantically under reparameterization; and avoid claiming a uniquely intrinsic constitutive component without evidence.
9. Compute hard structural target closure. If preserving a hard identity requires additional components to move, expand/retype the target or declare the proposed intervention inadmissible under that structural specification.
10. Audit intervention-reachable support and causal modularity. Identify every mechanism row that becomes relevant under J and state the evidence supporting stability of held-fixed mechanisms.
11. Build an acyclic rooted provenance graph for hard constraints, relevance choices, causal assumptions, support claims, model restrictions, and upstream certificates. Reject circular or self-validating support.
12. Construct the full provenance-valid identified model set M . Do not substitute one representative, average, majority, posterior-majority, or convenient model when the evidence leaves several models admissible.
13. Evaluate baseline and intervened path laws under the same frozen descriptor, relevance map, property map, and protocol for every required $m \in M$ and $y \in B_1$. Compute $\Delta_{\psi, J}(y)$ and, when uniform status is claimed, $\kappa_{\psi, J}(B_1)$. If the analysis uses approximated or estimated property outputs, propagate a justified two-sided error envelope and report whether κ exceeds the full baseline-plus-intervention error budget.
14. Assign protocol-scoped substantive status. Report robust PR, sign/direction, and uniform-gap status separately; report `ROBUST_NONCONSTITUTIVE_UNDER_FROZEN_PROTOCOL` for a negative result under one J ; reserve a family-wide absence or equivalence claim for an exhausted predeclared intervention family or theorem. For family comparisons, freeze and audit the full intervention signature rather than a favorable subset of labels.
15. When quantities are estimated, build one outer joint structural/statistical uncertainty set with justified simultaneous coverage for every dimension used by the claim. Account for multiple interventions, model selection, data-dependent search, and sequential monitoring/optional stopping as applicable.
16. If the valid outer set still contains a zero-constitution case, report `STATISTICALLY_UNRESOLVED` rather than PR or not-PR. Non-significance does not establish nonconstitution.
17. Emit the orthogonal application-certificate tuple: validity, epistemic mode, selection status, identification status, statistical status, provenance status, substantive protocol-scoped outcome, and uniform-gap status.
18. Preserve exploratory work as exploratory. Candidate regimes, relevance maps, model restrictions, or interventions discovered on development data require held-out or otherwise independent validation before being treated as confirmatory empirical results.

17. Appendix D - Formal Verification and Reproducibility

17.1 Verification architecture

The verification record has two deliberately separate layers. The first is executable and destructive: a frozen regression archive exercises historical controls, mutation cases, finite counterexamples, boundaries, pathologies, and representation failures. Its role is to expose mismatches between the manuscript contract and executable finite constructions. Passing those tests does not prove a universal theorem.

The second layer is deductive. The companion repository <https://github.com/HMarcusWH/PermanssonLean> formalizes the discrete mathematical core in Lean 4 against pinned mathlib dependencies. Its CI root build performs an axiom audit, runs `lake build`, and rejects `sorry`, `admit`, and source-level custom axiom declarations in the project source tree. The Lean development therefore checks the encoded theorem statements and their dependency chain against Lean's trusted kernel. This is a statement about the formalized mathematics, not about empirical truth, identification, provenance quality, or novelty.

17.2 Machine-checked scope

The discrete GR/PR mathematical core through Proposition 8.1 has corresponding Lean closure. The machine-checked scope includes joint-process construction and uniqueness; enlarged-state reduction; Polish descriptor-space support; exact and killed-kernel persistence; QSD-certified quasi-regimes; generalized and uniform Permansson constitution; perturbation robustness; the finite-horizon TV envelope; baseline, intervention-family, and quotient preservation; Paper-I world-only, recorded-action, and grounded transport; the Section 7 non-identification and nuisance-padding results; and the explicit period-two Exact GR witness.

This formalization scope is intentionally narrower than the full manuscript. The specialized finite-certificate and rigidity material in Appendix E is not part of the formalized universal theorem set. Continuous-time, differential-inclusion, nonautonomous, and application-specific extensions also remain outside the discrete-core claim. The application disciplines concerning relevance, causal modularity, support, provenance, identification, and statistical uncertainty also remain separate from the pure formal derivations.

17.3 Paper-to-Lean crosswalk

Table 8: Primary paper-to-Lean verification crosswalk. The repository’s docs/FORMALIZATION_MAP.md gives the complete dependency ledger.

Paper item	Mathematical object	Primary Lean closure
Thm. 3.1	Joint-process well-posedness	joint_process_well_posed
Prop. 3.2	Enlarged-state reduction	enlarged_state_reduction
Prop. 4.1a	Polish descriptor-space generality	proposition_4_1a
Prop. 4.2 / Thm. 4.2a	Exact invariance and killed persistence	exactInvariant_iff_survivalForever; survivalProbability_eq_killedSurvivalMass
Def. 4.4a / Prop. 4.4b	QSD-certified quasi-regime	IsQSDCertifiedQuasiRegime; proposition_4_4b
Prop. 5.2 / Cor. 5.3	Constitutive-margin robustness	constitutiveMargin_perturbation_abs_le; robustUniformPRCertificate
Prop. 5.4	Finite-horizon kernel-to-path TV envelope	proposition_5_4
Thm. 6.1	EGR embedding	paperI_EGR_iff_embeddedExactGR
Thm. 6.2	Conservative Paper-I recovery	paperIPermansson_iff_embeddedRelativePR plus recorded-action and grounded analogues
Prop. 7.1–7.4a	Representation and intervention-family invariance	exactGR_iff_of_inducedKernel_eq; frozenFamilyPR_iff_of_signature_match
Thm. 7.4b / Prop. 7.5	Quotient preservation and nuisance-padding exclusion	quotient_frozenFamilyPR_iff; nuisancePadding_cannot_create_grounded_change
Prop. 8.1	Periodic Exact GR	PeriodicExactGR.proposition_8_1

17.4 Reproducibility record

The formalization is pinned to Lean 4.34.0 and mathlib v4.34.0. The immutable formalization snapshot used for this manuscript is commit c018f79ea4ce46f4f679ad5bca254509778fc53c, whose root Lean workflow completed successfully in CI run 35808810682. The accompanying verification manifest records SHA-256 hashes for the final PDF, the final .tex source, and the frozen v0.1.7 post-proofread test archive. This creates a bidirectional provenance chain: the paper identifies the exact formalization commit, and the repository-side manifest identifies the exact paper and destructive-test artifacts associated with the release.

17.5 Revision and artifact lineage

The release history is retained here for reproducibility rather than as part of the theory exposition. The v0.1.6 manuscript serves as the earlier control lineage for the generalized framework and its initial formalization record. The v0.1.7 line adds the quantitative and representation-theoretic theorem layer and the associated destructive differential suite (Runs 17–30), covering killed-kernel indexing, QSD pathologies, noncompact descriptors, finite- versus infinite-horizon perturbations, constitutive-margin boundaries, intervention-family omissions, quotient failures, and the specialized research architectures described in Appendix E. The v0.1.8 release integrates the paper-to-Lean declaration crosswalk and an explicit release-verification gate over paper/source hashes, root-import reachability, CI evidence, figures, and artifact provenance.

The frozen v0.1.7 post-proofread destructive archive is preserved byte-for-byte as a historical verification artifact. Executable regression and mutation checks are falsification evidence rather than proof objects; the Lean development is the deductive verification layer. Exact artifact names, hashes, the immutable formalization commit, and the successful CI run are recorded in the accompanying release manifest and repository-side verification metadata.

17.6 Tool use and author responsibility

This is a working paper for academic circulation. Generative AI tools were used in literature triage, drafting, code generation, formal-consistency checks, adversarial validation, and language editing. The author reviewed the manuscript and retains responsibility for all claims. Computational checks and proof-assistant formalization supplement the manuscript exposition; they do not certify empirical inputs, causal identification, source provenance, statistical adequacy, or scholarly priority.

18. Appendix E - Specialized Finite-Certificate and Rigidity Extensions

The following theorem-transfer directions concern specialized subclasses with additional linear, bilinear, nested, or smooth structure that a general strategic-world model need not possess. They are therefore outside the universal GR/PR results established in the main text.

- **Scalar-defect track.** In a linear or quadratic subclass, agreement of two symmetric representations on a codimension-one core may confine their discrepancy to a rank-at-most-two channel, with an additional rigidity condition potentially collapsing it to a scalar rank-one common-mode defect. The finite Weil-CCM scalar-defect theorem supplies the proof architecture, not a universal nonlinear PR theorem (Hermansson, 2026b).
- **Persistent finite-detection track.** A strict full-object failure can transfer to a legal finite approximant when the failure functional is continuous in exactly the topology consumed by the certificate; persistent badness further requires value-preserving nested embeddings. Zero-margin failures, discontinuous functionals, or non-nested approximations do not qualify (Hermansson, 2026c).
- **First-bad reduction track.** If a natural Permansson certification hierarchy has a least failing level and successor quotient dimension r , the appropriate obstruction should generally be r -dimensional. Scalar secular reduction is licensed only when $r = 1$; it is not inferred merely from minimal failure (Hermansson, 2026d).
- **Singular-compatibility track.** When a future linearized response or predecessor operator is singular, exact analysis should begin with kernel/range compatibility rather than silently inserting an inverse or pseudoinverse. The companion zero-shift theorem motivates whole-kernel compatibility laws but does not itself establish them for arbitrary GR/PR models (Hermansson, 2026e).
- **Constructive quadratic witness track.** If a specialized constitutive comparison genuinely reduces to a signed rank-one difference $c(xx^\top - yy^\top)$ in a Hilbert space, the explicit Gram-witness theorem supplies a closed-form separating direction and perturbation margin. This remains a quadratic-response subclass, not a general representation of ψ (Hermansson, 2026f).

Scope condition. A specialized result belongs to the general framework only when a natural Permansson subclass supplies its hypotheses and a direct theorem establishes the claimed transfer. Analogy alone does not establish such an extension.

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